



Mechanical Unit Operations (CHL2050)

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Unit Operation: A unit operation is a fundamental physical process used in chemical and process industries to transform, transport, or separate materials without changing their chemical identity.

- It is a **physical change**.
- No new substance is formed.

Example: Sugarcane → Crushing → Filtration → Evaporation → Crystallization → Centrifugation → Drying

- Crushing → Size reduction
- Filtration → Separation
- Evaporation → Heat transfer
- Crystallization → Mass transfer
- Drying → Heat and mass transfer

Why don't we study complete industries separately?

Unit Operation	Unit Process
Physical change	Chemical change
No new substance formed	New substance formed
Filtration	Neutralization
Distillation	Oxidation
Drying	Polymerization
Crushing	Nitration

Unit Operation = Physical change

Unit Process = Chemical reaction

Why is it called *Unit* Operation? Why not just Operation?

A unit operation is a single, fundamental physical operation that **performs one specific function** in a manufacturing process. Several unit operations connected together form an entire chemical plant.

Why is this course called *Mechanical Unit Operations*?

The operations in which the **dominant force is mechanical** rather than thermal or chemical.

Operation	Main driving force	Category
Crushing rocks	Mechanical force	Mechanical Unit Operation
Grinding coal	Mechanical force	Mechanical Unit Operation
Screening sand	Mechanical motion	Mechanical Unit Operation
Filtration	Pressure (mechanical force)	Mechanical Unit Operation
Sedimentation	Gravity (mechanical force)	Mechanical Unit Operation
Pumping water	Mechanical energy	Mechanical Unit Operation

Mechanical Unit Operations	Thermal Unit Operations
Crushing	Evaporation
Grinding	Distillation
Screening	Drying
Filtration	Condensation
Sedimentation	Crystallization (often heat-assisted)

Contents

Particulate matter: Introduction, Particulate matter, size reduction, enlargement and separation, Crushers and Grinders, Sieve, Mixing, Agitation

Packed Columns and Granular Beds: Motion of Particles in a Fluid, flow through beds and columns, Single Fluid flow through Granular Beds, Specific Surface and Voidage, Packings, pressure drop in packed column,

Filtration and Membrane Separation: Theory, filtration equipment, membrane processes, general equation, microfiltration, fouling, pervaporation

Flotation and sedimentation: Fundamental concept, froth flotation, gravity settling, centrifugation, design criteria

Particulate Matter

What is Particulate Matter?

Particulate matter refers to systems containing **solid particles** — ranging from fine powders to coarse granules.

How is it Different from Fluids (Liquids & Gases)?

Feature	Fluids (Liquids/Gases)	Particulate Matter (Solids)
State Definition	Specified by T, P, & composition	Cannot be defined by T & P alone
Physical Properties	Fixed (density, viscosity, conductivity)	Variable; depends on particle arrangement
Key Characteristics	Uniform molecular behavior	Need particle size, shape, & composition
Geometrical Complexity	Simple continuous medium	Complex, irregular arrangements possible
Behavior Prediction	Precise equations available	Often requires empirical corrections

Storage types (solids)



Hoppers



Silos



Bins

Particulate Matter

Operation	Equipment / Method	Objective
Storage	Hoppers, Bins, Silos	Hold bulk solids; ensure controlled discharge without arching/rat-holing
Flow	Orifices, Pipes	Transport solids; predict flow rate
Metering	Screw feeders, Belt feeders, Rotary valves	Accurately dose & control solid flow rate
Size Reduction	Crushers (Jaw, Gyratory), Grinders (Ball mill, Hammer mill)	Break large particles into smaller, desired size
Agglomeration / Sintering	Pelletizers, Briquetting machines, Sintering furnaces	Form fine particles into larger aggregates or bonded masses
Mixing of Solids	Ribbon blender, V-blender, Paddle mixer	Achieve uniform distribution of two or more solid components
Separation	Screens (size), Cyclones, Classifiers, Magnetic separators (type)	Split mixture by particle size or material composition

Particle measurement: Particle shape

The Fundamental Problem:

For **regular particles** (spheres, cubes) → Shape is easily defined by mathematical equations.

For **irregular particles** (sand grains, mica flakes, crushed minerals) → Shape is **not clear** and must be **arbitrarily defined**.

The Solution: Sphericity (Φs): It tells you how close a particle is to a sphere. A perfect sphere has a sphericity of 1.0.

$$\Phi_s = \frac{6/D_p}{s_p/v_p}$$

where D_p = nominal diameter of particle
 s_p = surface area of one particle
 v_p = volume of one particle

Particle Type	Sphericity (Φs)
Perfect sphere	1.0
Cube	1.0
Short cylinder	1.0
Rounded particles (by abrasion)	Up to 0.95
Most crushed materials	0.6 – 0.8

Case Study 1: Cement plant

This plant was crushing limestone in a **standard ball mill**.

Their problem? The mill was consuming too much power – electricity bills were crushing their profits. And the cement quality was inconsistent.

What did they find?

The particles were highly irregular – sphericity around **0.62**. Very jagged, like broken glass. These irregular particles had two bad effects:

1. **First** – they packed too tightly in the kiln. The air couldn't flow through properly, so the kiln had to run hotter and longer. More fuel.- **high energy consumption**
2. **Second** – they didn't flow smoothly through the separator. The plant had constant blockages.- **low throughput**

The solution? They added a **roller press** before the ball mill as a pre-grinder. The roller press produces particles that are more rounded – sphericity increased to **0.71**.

The result?

- Energy consumption dropped by **15%**
- Production increased by **10%**

The plant saved **₹2 crore per year**

All because they made the particles slightly more spherical.

Case Study 2: A pharmaceutical company making paracetamol tablets.

They were using **crystalline lactose** as the filler in their tablets. Lactose crystals are shaped like tiny stones – sharp edges, irregular. Sphericity around **0.65**.

The problem? The powder wouldn't flow smoothly into the **tablet press**. The machine would jam every 20 minutes. When tablets did come out, their weight varied by as much as **8%**. Too much medicine in one and too less medicine in other tablet.

The solution? They switched to **spray-dried lactose**. This is lactose that has been dissolved in water and then sprayed into hot air. The droplets dry into tiny **spheres** – sphericity close to **0.92**.

The result?

- Tablet weight variation dropped from 8% to **under 1%**
- Machine jams reduced by **90%**
- Production speed increased from 100,000 to **350,000 tablets per hour**

How to Determine Sphericity Experimentally:

Method	When to Use
Pressure drop in a packed bed	For granular materials
Adsorption measurements	For non-porous particles

Why Does Particle Shape Matter in Operations?

Operation	Effect of Irregular Shape
Flow through hoppers	Bridging, arching, rat-holing
Size reduction	Different fracture patterns
Mixing	Segregation tendency
Filtration / Packed beds	Affects void space & pressure drop
Sedimentation	Alters drag force & settling velocity

Particle Size

What is Particle Size?

- For **equidimensional particles** (roughly same in all directions) → "Diameter" can be specified
- For **non-equidimensional particles** (longer in one direction) → Characterized by the **second-longest major dimension**

Units by Particle Size Range

Size Range	Typical Units	Example
Coarse particles	Inches (in), Millimeters (mm)	Gravel, crushed stone
Fine particles	Screen size (mesh number)	Sand, flour
Very fine particles	Micrometers (µm), Nanometers (nm)	Cement powder, pigments
Ultrafine particles	Surface area per unit mass (m²/g)	Nanoparticles, activated carbon

Particle Size Distribution (PSD)

- Industrial powders contain particles of different sizes.
- PSD describes the proportion of particles present in each size range.
- Determined by separating particles into size fractions and measuring mass or number in each fraction.

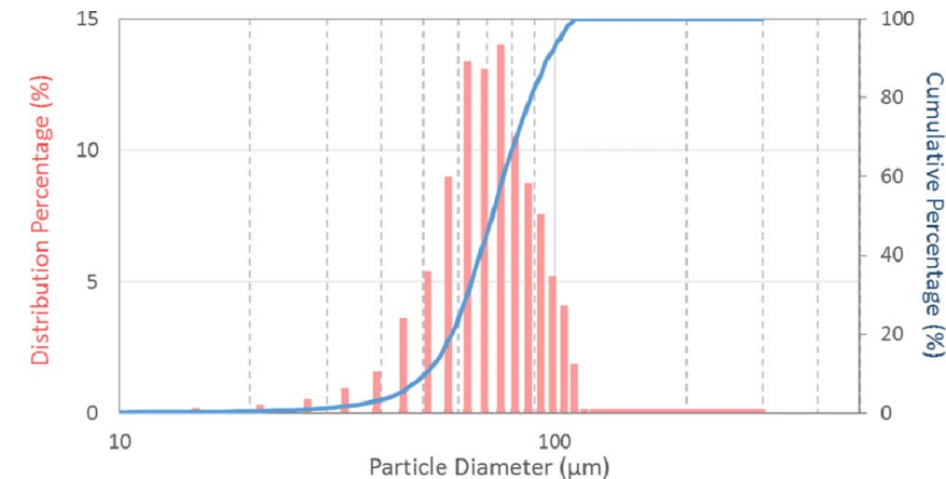
Methods of Presenting PSD (Histogram)

1. Differential Analysis

1. Mass fraction plotted against particle size interval.
2. Usually shown as a histogram.
3. Indicates where most particles are concentrated.

2. Cumulative Analysis

1. Shows percentage of particles smaller than a given size.
2. Obtained by summing differential fractions.



Case Study : Cement Manufacturing.

- In a cement plant, limestone is ground into fine powder. The final product must have particles mostly between **1 and 100 micrometers**.
- If particles are too coarse – say above 100 microns – the cement won't hydrate fully. The building will be weak.
- If particles are too fine – below 1 micron – the cement becomes too thirsty for water. The concrete cracks.
- So how to control this? Every hour, we take a sample and runs it through a **laser diffraction particle size analyzer**. The machine gives a full size distribution in 2 minutes. If the distribution shifts, they adjust the grinding mill immediately.
- A cement plant could reduce off-spec product from 8% to 2% simply by installing online particle size monitoring. That saved them crores of rupees per year.

Mixed Particle Sizes & Key Equations

Number of particles in a uniform sample.

$$N = \frac{m}{\rho_p v_p}$$

Total surface area.

$$A = \frac{6m}{\Phi_s \rho_p D_p}$$

Term	Symbol	Units
Mass of sample	m	kg, g, mg
Particle density	ρ_p	kg/m ³ , g/cm ³
Volume of one particle	v_p	m ³ , cm ³
Number of particles	N	dimensionless
Surface area of one particle	S_p	m ² , cm ²
Total surface area	A	m ² , cm ²
Sphericity	Φ_s	dimensionless
Nominal diameter	D_p	m, cm, mm, μm

Case Study 1: Asthma Inhalers.

- A pharmaceutical company makes an inhaler for asthma patients. The active drug particles must be between **1 and 5 microns**.
- **Why?** Because particles larger than 5 microns get stuck in the throat and don't reach the lungs. Particles smaller than 1 micron get exhaled out.
- **The company needs to know:** How many drug particles are in each puff? The dose is measured in micrograms, but the lung responds to particle count.
- Using the number of particles equation, they calculate that for **100 micrograms of drug with average diameter 3 microns** and $\alpha = 0.52$ (spherical shape), the number of particles is approximately 2 billion per puff.
- If the size drifts to 4 microns, the number of particles drops by more than half. The patient gets less medicine. That's not acceptable.

Case Study 2: Pharmaceutical Tablets.

When you make a tablet, the active drug particle size must be controlled very precisely.

Why: Two reasons:

1. **dissolution rate.** Smaller particles dissolve faster. If paracetamol particles are too large, the tablet won't work quickly enough when a patient takes it. If they are too small, the drug might release too fast and cause side effects.
 2. **content uniformity.** Suppose tablet has 500 mg of paracetamol. You need exactly 500 mg in every tablet. If the particle size varies too much, some tablets will have more, some less.
- A major pharma company found that reducing the active ingredient's average particle size from 50 microns to 20 microns improved dissolution rate by 40%. But they also had to control the distribution tightly – 90% of particles had to be between 15 and 25 microns.
 - They achieved this using ***micronization – a high-energy milling process.***
 - **The result?** A better product and faster FDA approval

Specific Surface of a Mixture (Aw)

Specific surface (Aw) = Total surface area of particles **per unit mass** of sample. Units: m²/g or cm²/g

For a Single Size Fraction (from previous slide):

$$A = \frac{6m}{\Phi_s \rho_p D_p}$$

For a Mixture (Multiple Sizes):

$$A_w = \frac{6}{\Phi_s \rho_p} \left(\frac{x_1}{D_{p1}} + \frac{x_2}{D_{p2}} + \dots + \frac{x_n}{D_{pn}} \right) = \frac{6}{\Phi_s \rho_p} \sum_{i=1}^n \frac{x_i}{D_{pi}}$$

Key Assumptions:

Assumption	Meaning
Φ_s (sphericity) is constant across all fractions	Same shape for all particles
ρ_p (particle density) is constant across all fractions	Same material for all particles

Average Particle Sizes

Name	Symbol	Formula	When to Use
Volume-Surface mean diameter (Sauter mean)	D_s	$\frac{1}{\sum (x_i/D_{pi})}$	Counting pollen grains under a microscope, measuring bacteria or blood cells where each particle is equally important.
Arithmetic mean diameter	D_N	$\frac{\sum (N_i D_{pi})}{N_T}$	Determining the average size of cement, sugar, or fertilizer particles in a bag where the amount (mass) of material matters.
Mass mean diameter	D	$\sum (x_i D_{pi})$	Measuring droplet sizes in milk, paint, or pharmaceutical suspensions using laser diffraction, where larger particles contribute more to the total volume.
Volume mean diameter	D_v	$\left(\sum \frac{x_i}{D_{pi}^3} \right)^{-1/3}$	Designing a fluidized bed reactor, spray dryer, or catalyst bed, where the available surface area controls the rate of heat or mass transfer.

Volume Shape Factor

Volume Shape Factor (α): Volume of a particle = $\alpha \times (\text{Diameter})^3$ $v_p = \alpha D_p^3$

Particle Shape	α (Volume Shape Factor)	Sphericity (Φ_s)
Sphere	0.5236 ($\pi/6$)	1.0
Short cylinder (height = diameter)	0.785 ($\pi/4$)	1.0
Cube	1.0	1.0

Size Reduction

Definition: Mechanical process of cutting or breaking solid particles into smaller pieces.

Industrial Objectives:

- Produce workable size (e.g., crushing crude ore).
- Achieve powder form (e.g., grinding synthetic chemicals).
- Create specific shapes (e.g., cutting plastic into cubes/diamonds).

Why Reduce Size?

- **Meet Specs:** Products require stringent particle size/shape standards.
- **Increase Reactivity:** Smaller particles = higher surface area = faster chemical reactions.
- **Enable Separation:** Allows mechanical liberation of unwanted ingredients.
- **Reduce Bulk:** Makes fibrous materials easier to handle and dispose

Size Reduction

Mode	Mechanism	Analogy	Application
Compression	Slow, heavy pressure	Nutcracker	Coarse reduction of hard solids (few fines)
Impact	Instantaneous, forceful blow	Hammer	Coarse, medium, or fine products
Attrition (Rubbing)	Shearing/abrasion between surfaces	File	Very fine products from soft, non-abrasive materials
Cutting	Sharp blade shearing action	Shears	Definite particle size/shape (few fines)

Energy & Power Requirements in Size Reduction

The Fracture Mechanism (What happens inside the particle):

Stage	Process	Energy Fate
1. Strain	Particles distort & stretch	Energy stored as mechanical stress (like a spring)
2. Rupture	Ultimate strength exceeded → particle breaks	Stored energy released
3. New Surface	Fragments form	Energy used to create surface energy
4. Heat	Excess energy dissipates	Lost as heat (by conservation of energy)

Crushing laws: Rittinger's law

Three Classical Energy Laws

- ◆ Rittinger's Law (1867)
- ◆ Kick's Law (1885)
- ◆ Bond's Law (1952)

Coarse Crushing → Intermediate Grinding → Fine Grinding

Kick → Bond → Rittinger

Rittinger's Law: Rittinger's Surface Area Theory

$$E = K_R f_c \left(\frac{1}{L_2} - \frac{1}{L_1} \right)$$

Assumption: Energy consumed is proportional to: New Surface Area Created

Interpretation

- Smaller particles possess larger surface area.
- Most energy is used in creating fresh surfaces.
- Applicable for **fine grinding**.

Best Used For: Ball mills, Ultrafine grinding, and Cement grinding

Limitation: Overpredicts energy for coarse crushing.

Crushing laws: Kick's law

Kick's Law: Kick's Reduction Ratio Theory

*Power cost is a major expense in crushing & grinding operation

The Fracture Mechanism (What happens inside the particle)

$$E = K_K f_c \ln \left(\frac{L_1}{L_2} \right)$$

Assumption: Energy depends only on the reduction ratio:

$$\frac{L_1}{L_2}$$

Example: 50 mm → 25 mm; Reduction ratio = 2

12 mm → 6 mm; Reduction ratio = 2

According to Kick: → Both operations require approximately the same energy.

Best Used For: Coarse crushing, Jaw crushers, Gyratory crushers

Physical Meaning: Energy is mainly used for elastic deformation before fracture.

Symbol	Description	Unit
E	Energy required for size reduction per unit mass	kWh/ton, J/kg
L1	Initial particle size (feed size)	mm, μm
L2	Final particle size (product size)	mm, μm
C	Empirical constant	Depends on law
P	Exponent in general energy equation	Dimensionless
fc	Crushing strength of material	N/m² or MPa
KR	Rittinger's constant	Material dependent
KK	Kick's constant	Material dependent
(Ei	Bond Work Index	kWh/ton
(q	Reduction ratio ((L1/L2))	Dimensionless

Crushing laws: Bond's law & Work Index

Why Bond's Theory?

- Rittinger's law works best for **fine grinding**.
- Kick's law works best for **coarse crushing**.
- Neither accurately predicts energy requirements over a wide size range.
- **Bond proposed an intermediate theory**, widely used in industrial comminution design.

Bond's Crushing Law

Bond assumed: Energy required is proportional to the square root of the new surface-to-volume ratio created during size reduction.

Energy-Power Relationship

$$\frac{P}{m} = \frac{K_B}{\sqrt{D_p}}$$

Symbol	Meaning
(P)	Power consumed (kW)
(m)	Throughput (ton/h)
(Dp)	Product particle size (mm)
(KB)	Bond constant

Crushing laws: Bond’s law & Work Index

Bond Work Index (W_i): **Work Index (W_i)** is the energy required to reduce **1 ton of material** from a very large feed size to a product in which **80% passes through a 100 μm screen**.

Units: W_i (kWh/ton)

Physical Meaning

- Measure of material hardness/grindability.
- Higher $W_i \rightarrow$ More difficult to grind.
- Lower $W_i \rightarrow$ Easier to grind.

Symbol

Dpa

Dpb

Wi

P/m

Meaning

Feed size at 80% passing (mm)

Product size at 80% passing (mm)

Bond Work Index (kWh/ton)

Specific energy consumption

Practical Bond Equation:

$$\frac{P}{m} = 0.3162W_i \left(\frac{1}{\sqrt{D_{pb}}} - \frac{1}{\sqrt{D_{pa}}} \right)$$

Crushing laws

EXAMPLE 28.2. What is the power required to crush 100 ton/h of limestone if 80 percent of the feed passes a 2-in. screen and 80 percent of the product a $\frac{1}{8}$ -in. screen?

Solution. From Table 28.2, the work index for limestone is 12.74. Other quantities for substitution into Eq. (28.17) are

$$\dot{m} = 100 \text{ ton/h}$$

$$D_{pa} = 2 \times 25.4 = 50.8 \text{ mm} \quad D_{pb} = 0.125 \times 25.4 = 3.175 \text{ mm}$$

The power required is

$$\begin{aligned} P &= 100 \times 0.3162 \times 12.74 \left(\frac{1}{\sqrt{3.175}} - \frac{1}{\sqrt{50.8}} \right) \\ &= 169.6 \text{ kW (227 hp)} \end{aligned}$$

Size reduction

Equipment for size reduction

Convert large solid particles into smaller particles for easier handling, processing, and reaction.

Classification of Size Reduction Equipment

Equipment	Feed Size	Product Size	Main Mechanism
Crusher	Large rocks (upto 1.5 m size)	6–250 mm	Compression
Grinder	Crushed solids (below 6 mm)	Powder (40 mesh screen)	Impact + Attrition
Ultrafine Grinder	Fine feed (<6 mm)	1–50 µm	Attrition
Cutter	Sheets, fibers, plastics	Definite shape (2–10 mm)	Cutting

Crushers and Their Types

A crusher is a **slow-speed machine** used for coarse reduction of large quantities of solids.

Major Types of Crushers (Compression- operated)

1. Jaw Crusher

- Feed enters between two jaws.
- One jaw is fixed.
- Other jaw moves back and forth.
- Material is crushed by compression.

Applications

- ✓ Primary crushing of rocks and ores
- ✓ Mining industry
- ✓ Cement plants

2. Gyratory Crusher

- Conical crushing head gyrates inside a casing.
- Material is repeatedly compressed and broken.
- Product exits from the bottom.

Advantages

- ✓ High capacity
- ✓ Continuous operation
- ✓ Suitable for very large feed sizes

3. Roll Crushers

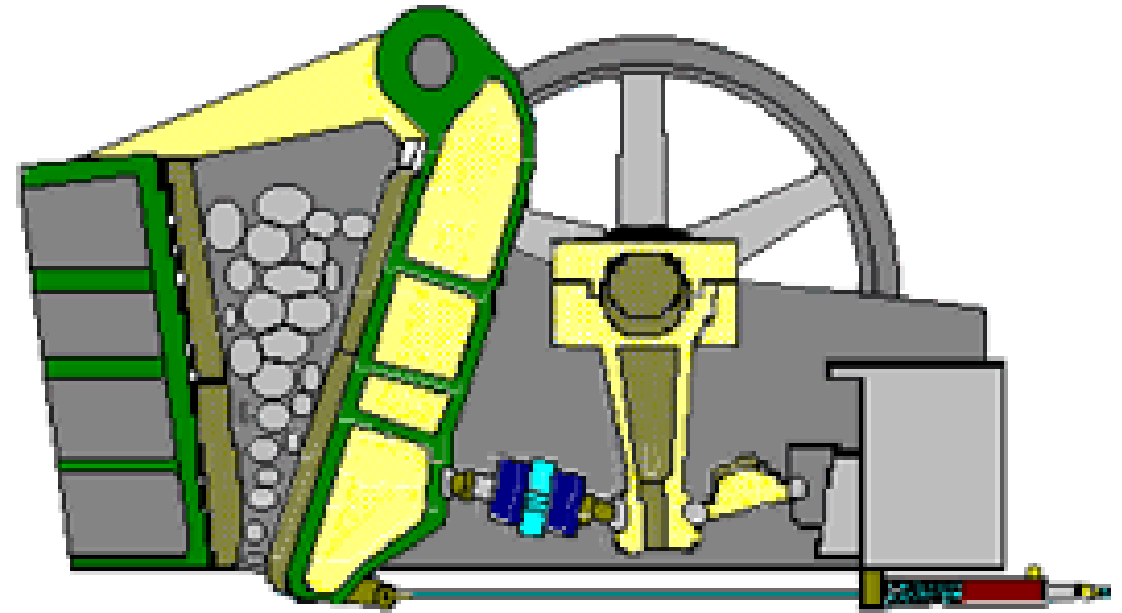
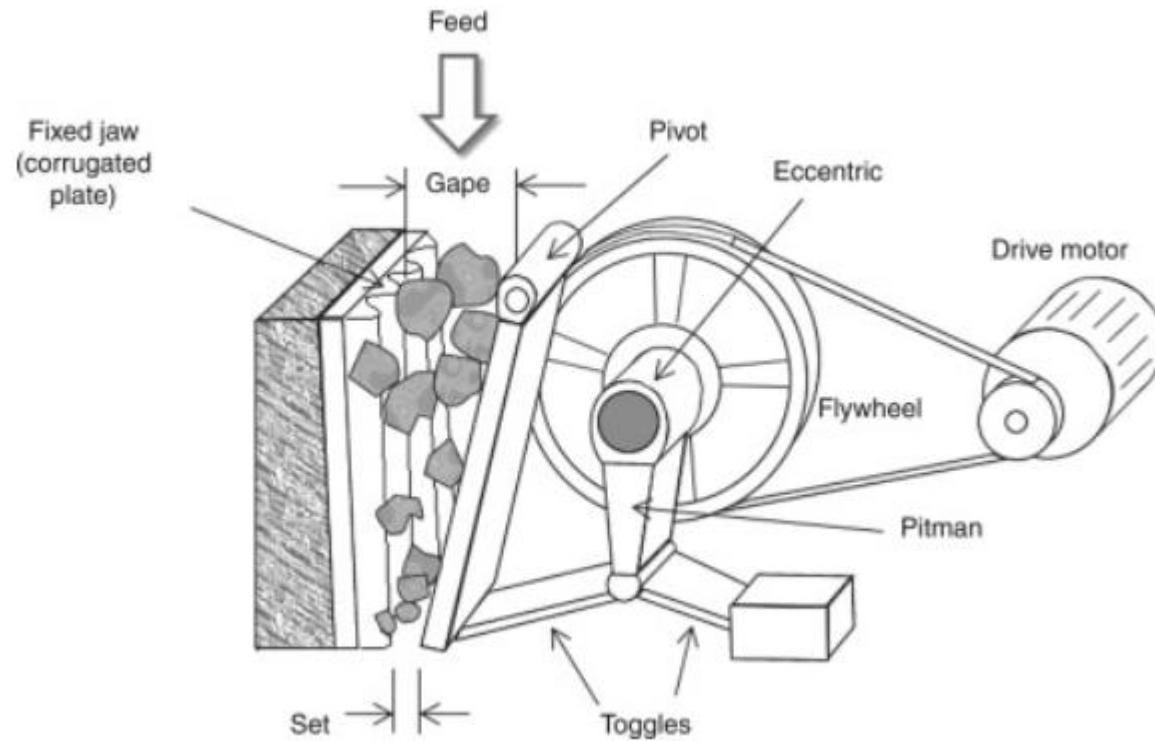
Smooth-Roll Crusher

- Two rotating rolls compress particles.
- Produces particles of about 1–12 mm.

Toothed-Roll Crusher

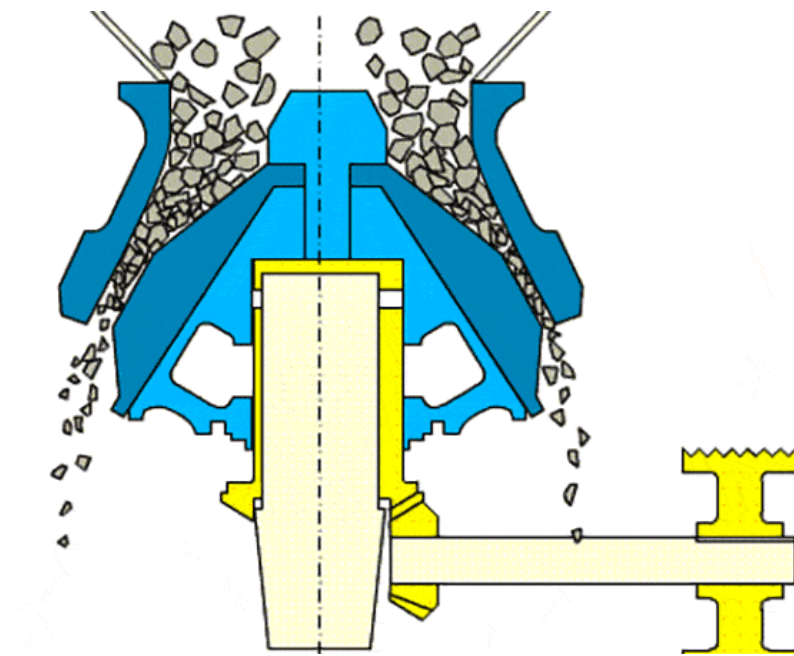
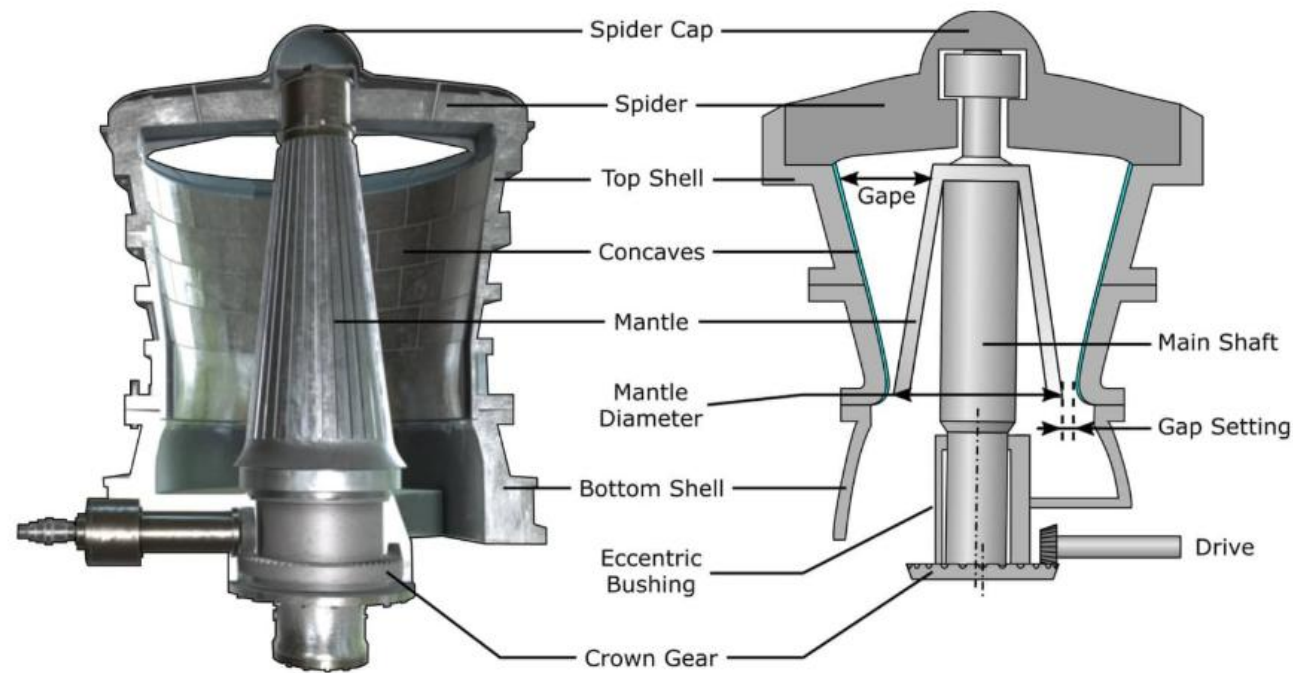
- Teeth grip and pull material into rolls.
- Suitable for brittle materials such as coal

Jaw Crusher



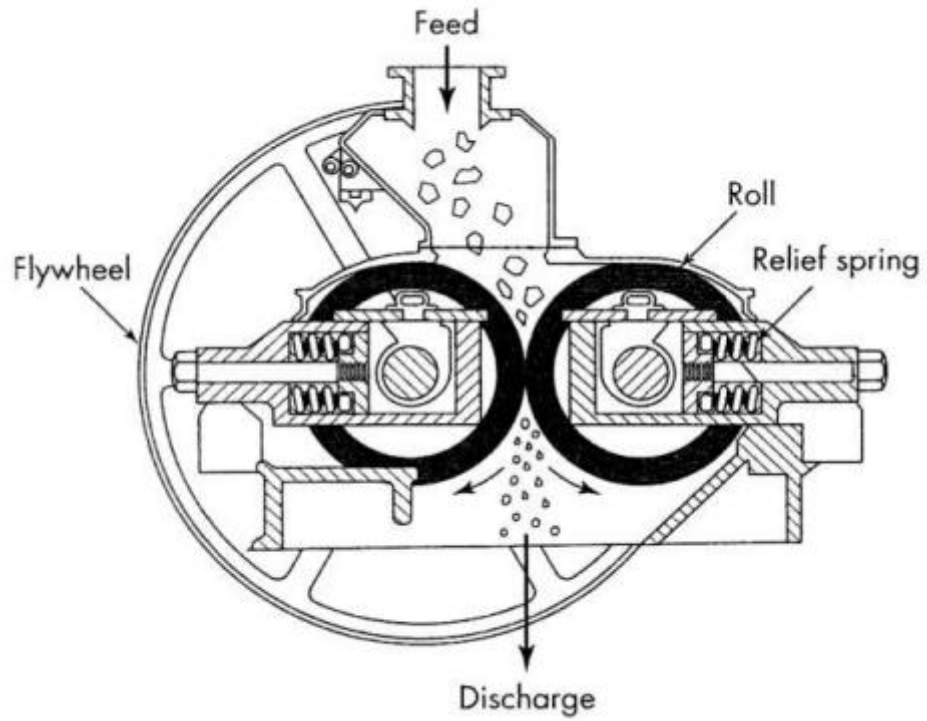
- The cycle repeats 200 to 300 times per minute.
- The product from a primary jaw crusher is 6 to 250 mm. The reduction ratio—the ratio of feed size to product size—is usually 3:1 to 6:1.

Gyratory crusher

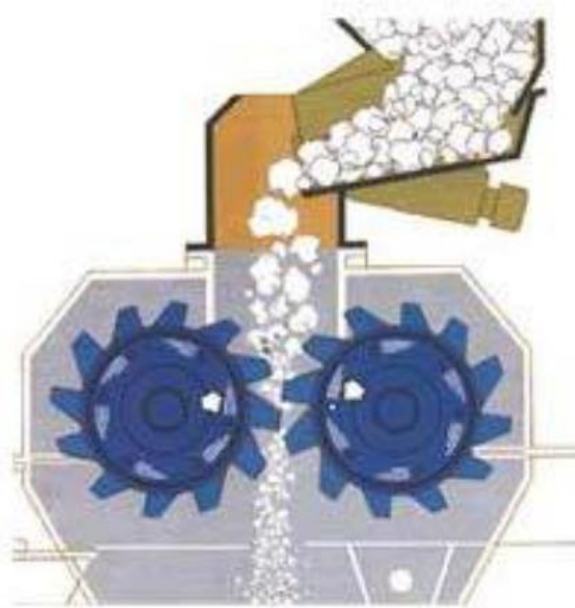


Size reduction

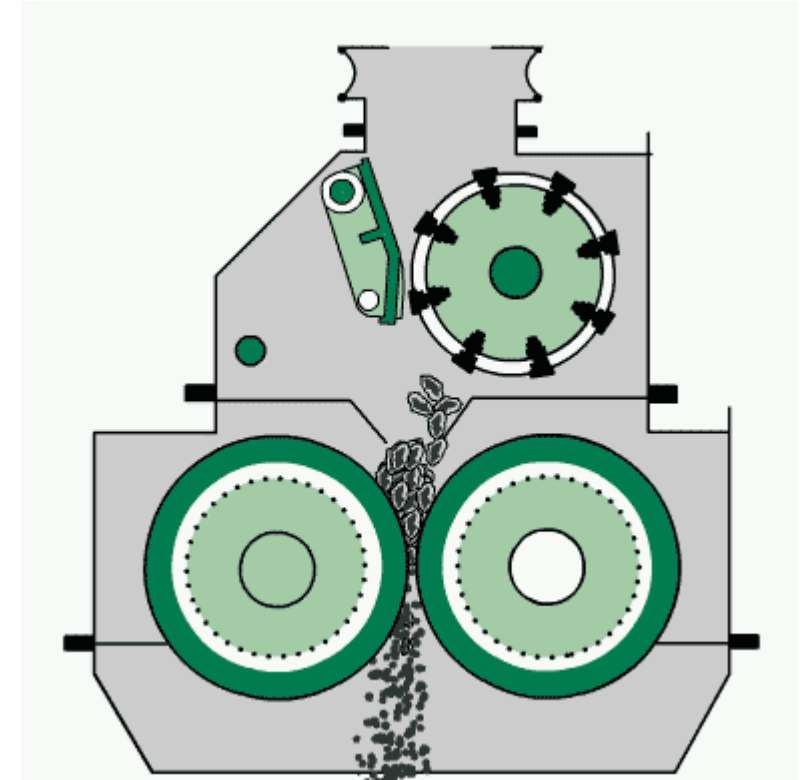
Roller crusher



Smooth Roller Crusher



Toothed Roller Crusher

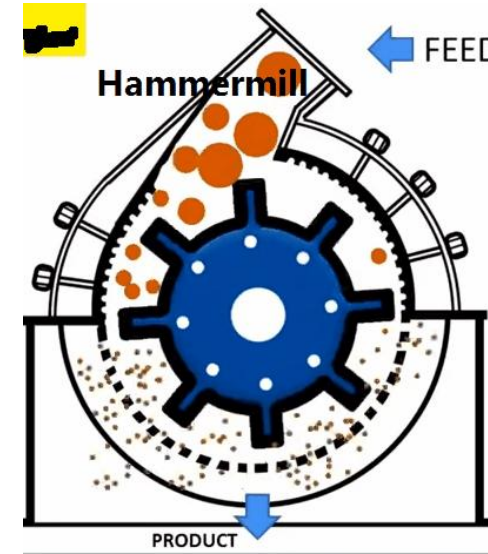
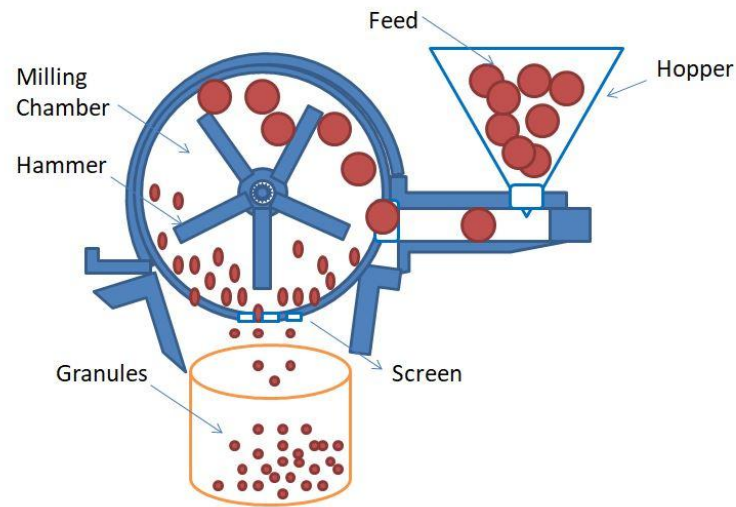


- It is strictly a **secondary crusher**
- Feed size should not exceed 5% of the roll diameter.

Grinders: Intermediate Size Reduction

- **Grinders accept product from a crusher for further reduction**
- Four main types of commercial grinders:
 1. Hammer mills and impactors
 2. Rolling-compression machines (roller mills)
 3. Attrition mills
 4. Tumbling mills

Hammer Mills

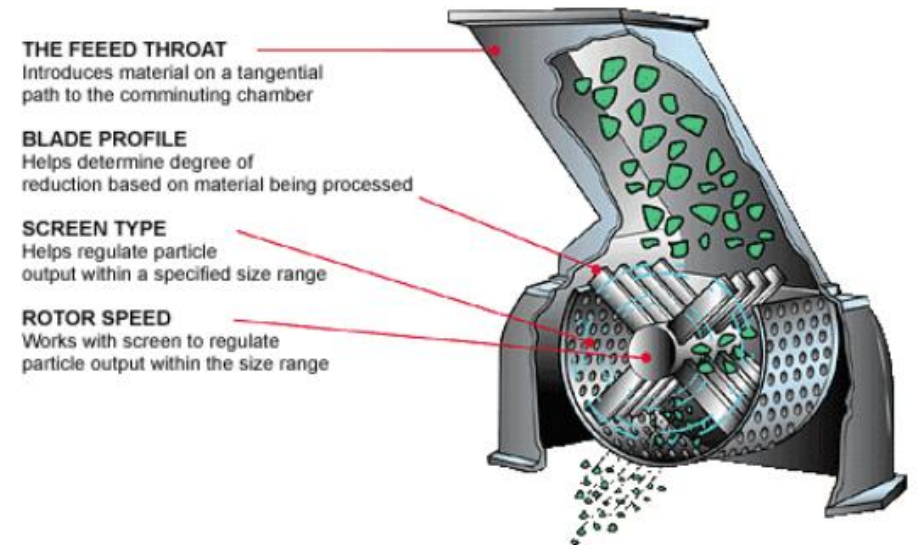


Hammer Mills – Construction & Mechanism

- **High-Speed Impact**
- Each disk carries four to eight swing hammers
- Hammers may be straight metal bars with plain or enlarged ends, or sharpened to a cutting edge
- Contains a high-speed rotor turning inside a cylindrical casing
- Feed dropped into the top of the casing is broken and falls out through a bottom opening
- Particles are broken by sets of swing hammers pinned to a rotor disk
- Fragments are rubbed into powder by the hammers and pushed through a grate or screen
- Several rotor disks (150 to 450 mm diameter) often mounted on same shaft

Hammer Mill Capabilities

- Fine reduction hammer mills: peripheral **speed of hammer tips may reach 110 m/s** (360 ft/s)
- Fine mills reduce **0.1 to 15 tons/hour** to sizes finer than 200-mesh
- Hammer mills grind almost anything:
 - Tough fibrous solids (bark, leather)
 - Steel turnings
 - Soft wet pastes
 - Sticky clay
 - Hard rock
- **Limitation for fine reduction:** restricted to softer materials only
- Commercial mills typically reduce 60 to 240 kg of solid per kilowatt-hour (100 to 400 lb/hp·h)



Hammer Mills

Case Study 1: Hammer Mill in Corn Flour Production (Food Industry)

- A food processing company produces corn flour for breakfast cereals and snack products.
- Whole corn kernels must be reduced to fine particles before mixing and extrusion.
- Initially, the company used a **roller mill**, but frequent clogging occurred due to varying moisture content and the hard outer husk of the kernels.
- Engineers replaced the roller mill with a **hammer mill**, which uses high-speed rotating hammers to break the kernels by impact.
- A **500 µm screen** was installed to obtain a uniform flour particle size.
- Production capacity increased by **about 40%**, and maintenance costs decreased due to fewer blockages.
- However, operating the hammer mill at very high rotor speeds **generated excessive heat**, causing partial starch gelatinization and reducing flour quality.
- Engineers optimized the **rotor speed, feed rate, and screen size** to maintain high productivity while preventing overheating.

Case Study 2: Hammer Mill in Pharmaceutical Powder Production

- A pharmaceutical company manufactures tablets containing a **heat-sensitive active pharmaceutical ingredient (API)**.
- Before blending and tableting, the API and excipients must be reduced to **less than 250 μm** .
- A **hammer mill** was selected because it provides rapid and uniform particle size reduction.
- During large-scale production, repeated impact between hammers and particles generated excessive heat inside the milling chamber.
- The temperature rise caused partial degradation of the API, reducing drug potency and product quality.
- Engineers reduced the **rotor speed**, introduced **intermittent feeding**, and installed an **air-cooling system** around the mill.
- These modifications maintained the required particle size while protecting the heat-sensitive drug.

Case Study 3: Hammer Mill in Plastic Recycling

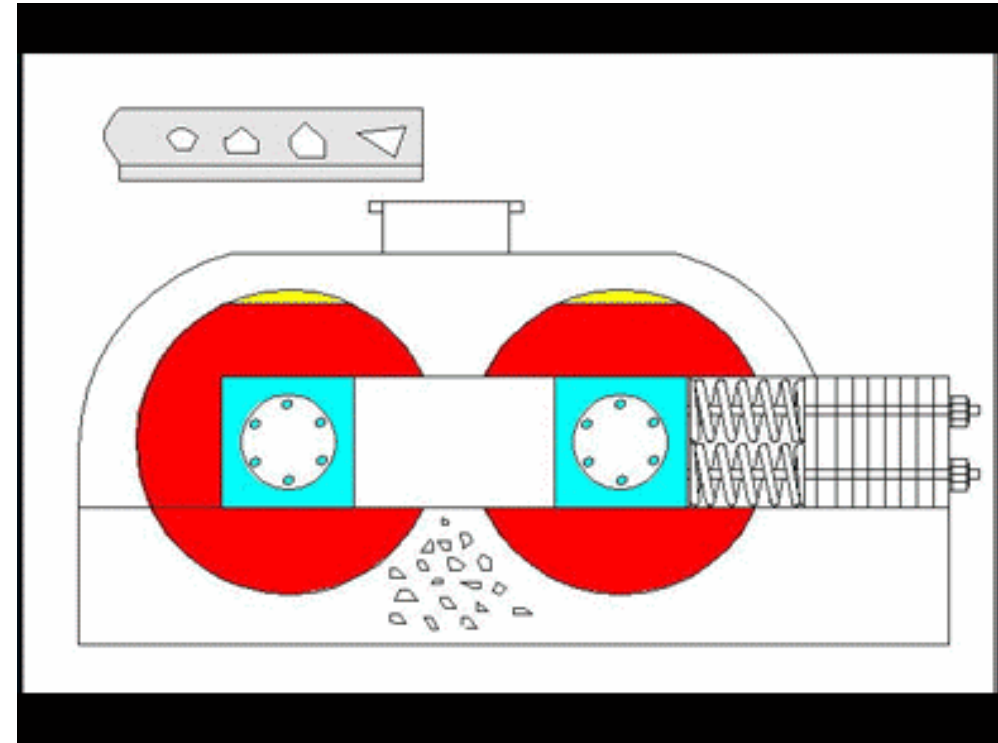
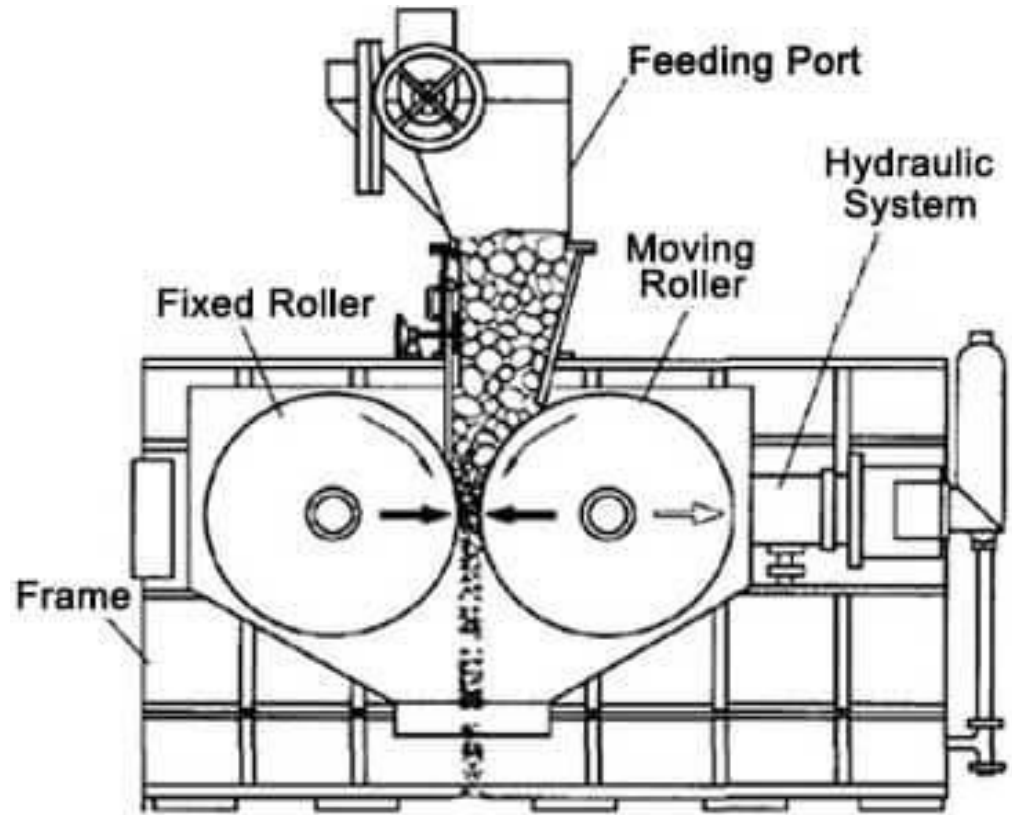
- A plastic recycling plant processes discarded **HDPE and PP containers** before washing and extrusion.
- The plastic waste must be reduced into flakes of approximately **8–10 mm**.
- A **hammer mill** was installed because it efficiently fractures rigid plastic containers and helps remove paper labels.
- The system worked well for bottles and rigid packaging materials.
- However, flexible plastic films and plastic bags wrapped around the rotating hammers, causing severe blockages and frequent shutdowns.
- Engineers installed a **pre-shredder** to separate and remove flexible plastics before hammer milling.
- The hammer mill was then used only for rigid plastics, significantly improving productivity and reducing maintenance downtime.

Roller Mills

- **Rolling Compression**
- Solids caught and crushed between vertical cylindrical rollers and a stationary anvil ring (bull ring)
- Product swept out of the mill by a **stream of air to a classifier separator**
- Bowl or ring is driven, rollers rotate on stationary axes (vertical or horizontal)
- Greatest application in reduction of:
 - Limestone
 - Cement clinker
 - Coal
- Pulverize up to 50 tons/hour

Size reduction

Roller Mills



Case Study: Coal Pulverization in a Thermal Power Plant

- A **500 MW coal-fired thermal power plant** uses a **vertical roller mill (VRM)** to prepare pulverized coal.
- Raw coal enters the mill with a particle size of **25–40 mm**.
- Heavy rollers apply **compressive forces** to grind the coal.
- **Hot air** simultaneously: Dries the coal & Carries fine particles toward a classifier.
- The classifier sends particles **smaller than 75 μm** to the boiler burners and Returns oversized particles to the grinding zone.
- **During the monsoon**, coal moisture increased from **8% to 15%**, causing: Reduced mill throughput & Increased energy consumption.
- Engineers corrected the problem by: Increasing the **hot air temperature** & Adjusting the **classifier speed**.
- These changes restored the required coal fineness and maintained boiler efficiency.

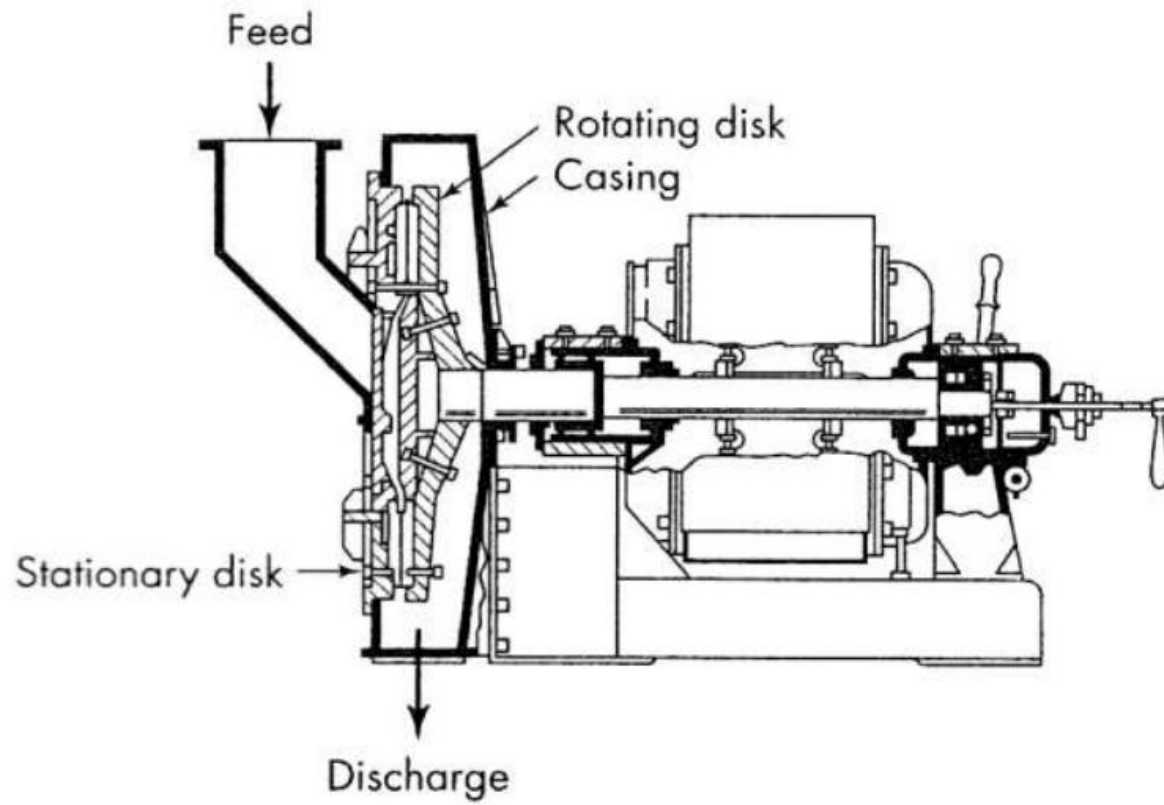
Attrition Mills

Attrition Mills: Soft Solids by Rubbing

- Particles of soft solids are rubbed between grooved flat faces of rotating circular disks
1. **Single-runner mill:** one disk stationary, one rotates
 2. **Double-runner machine:** both disks driven at high speed in opposite directions
- Feed enters through an opening in the hub of one disk
 - Suitable only for soft, non-abrasive materials
 - Produces very fine products primarily by attrition (rubbing) rather than impact

Size reduction

Attrition Mills



Attrition Mills

Case Study: Cocoa Powder Production Using an Attrition Mill

- A chocolate manufacturing plant produces **high-quality cocoa powder** for beverages and confectionery products.
- Roasted cocoa cake is first broken into coarse particles of **2–5 mm**.
- The material is fed into an **attrition mill**, where two rough grinding surfaces rotate relative to each other.
- Size reduction occurs mainly by **shearing and rubbing (attrition)** rather than impact.
- The mill produces cocoa powder with an average particle size of **20–50 µm**, providing a smooth mouthfeel.
- During production, engineers observed that increasing the rotor speed improved fineness but also raised the powder temperature due to friction.
- Excessive heating caused: Loss of cocoa aroma, Partial melting of cocoa butter, Reduced product quality.
- The plant installed a **water-cooled grinding system** and optimized the rotor speed to maintain product temperature.
- The optimized process resulted in: Uniform particle size, Better flavor retention, Lower energy consumption.

Tumbling Mills – Batch vs. Continuous Operation

- Unlike other mills requiring continuous feed, tumbling mills may operate in either batch or continuous mode

Batch Operation:

- Measured quantity of solid loaded into mill through an opening in the shell
- Opening closed, mill runs for several hours
- Mill stopped, product discharged

Continuous Operation: Solid flows steadily through the revolving shell

Basic Mechanism

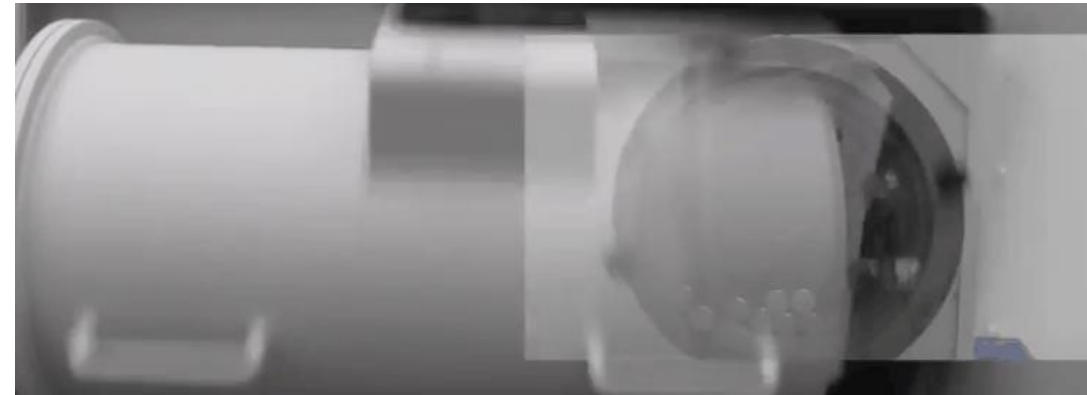
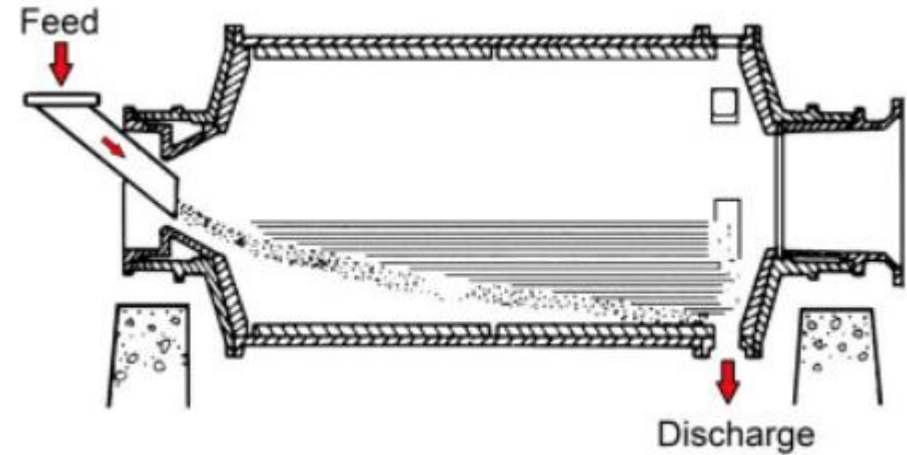
- Grinding media (balls/rods) are carried up the side of the shell nearly to the top
- Media then fall onto particles underneath
- Energy expended in lifting grinding units is utilized in particle size reduction

Tumbling Mills: Rod Mills

- **Rod Mills – Rolling Compression & Attrition**
- Much of the reduction done by rolling compression and attrition
- Rods slide downward and roll over one another
- Grinding rods are steel, **25 to 125 mm** (1 to 5 in.) in diameter
- Several rod sizes present at all times in any given mill

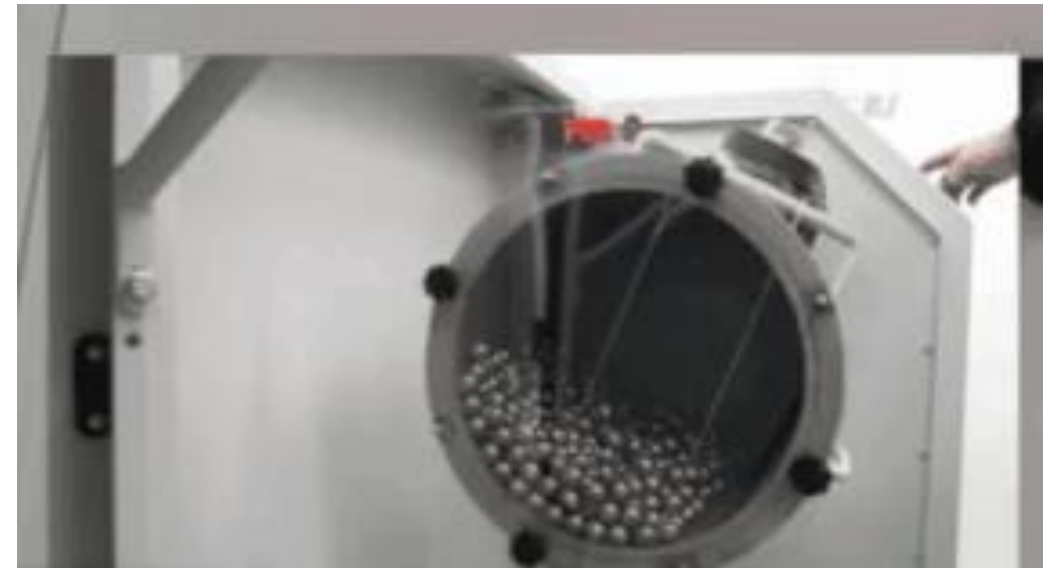
Application:

- Intermediate grinders
- **Reduce 20 mm feed to approximately 10-mesh**
- Often prepare crusher product for final reduction in a ball mill



Tumbling Mills: Ball Mills & Pebble Mills – Impact

- Ball Mills and Pebble Mills: **Impact Grinding**
- Most reduction done by impact as balls or pebbles drop from near the top of the shell
- **Typical Dimensions (Large Ball Mill):**
 - Shell diameter: 3 m (10 ft)
 - Shell length: 4.25 m (14 ft)
- **Grinding Media Sizes:**
 - Ball mill balls: 25 to 125 mm (1 to 5 in.) diameter
 - Pebble mill pebbles: 50 to 175 mm (2 to 7 in.) diameter
- Made up of Porcelain **or ceramic (zirconia mostly)**



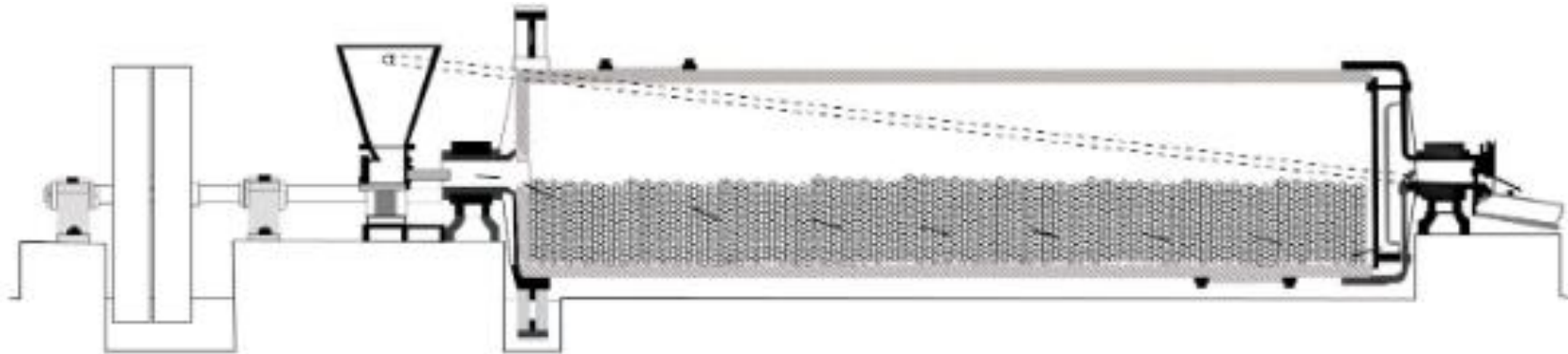
Tumbling Mills: Tube Mills

Tube Mills – Long Residence Time

- **Title:** Tube Mills: Extended Grinding (residence time 15 to 20 min)
- Continuous mill with a long cylindrical shell
- Material ground for 2 to 5 times as long as in a shorter ball mill

Key Characteristic:

- Excellent for grinding to very fine powders in a single pass
- Used where energy consumption is not of primary importance



Case Study: Ceramic Tile Manufacturing

- A ceramic tile plant produces **500 tonnes of floor and wall tiles per day**.
- Raw materials such as **clay, feldspar, quartz, and silica sand** are mixed with water.
- The mixture is fed into a **wet tube mill** containing alumina or ceramic grinding media.
- As the mill rotates, the grinding media tumble and reduce the particle size through **impact and attrition**.
- Grinding continues until the slurry reaches an average particle size of **10–20 µm**.
- The finely ground slurry (slip) is then pumped to a **spray dryer** to produce granules for tile pressing.
- Engineers observed that increasing the milling time beyond the optimum value produced only a slight reduction in particle size while significantly increasing energy consumption.
- By optimizing the milling time and grinding media loading, the plant achieved:
 - Lower electricity consumption.
 - Uniform particle-size distribution.
 - Improved tile strength and surface finish.
 - Reduced rejection during firing.

Tumbling Mills: Compartment Mills

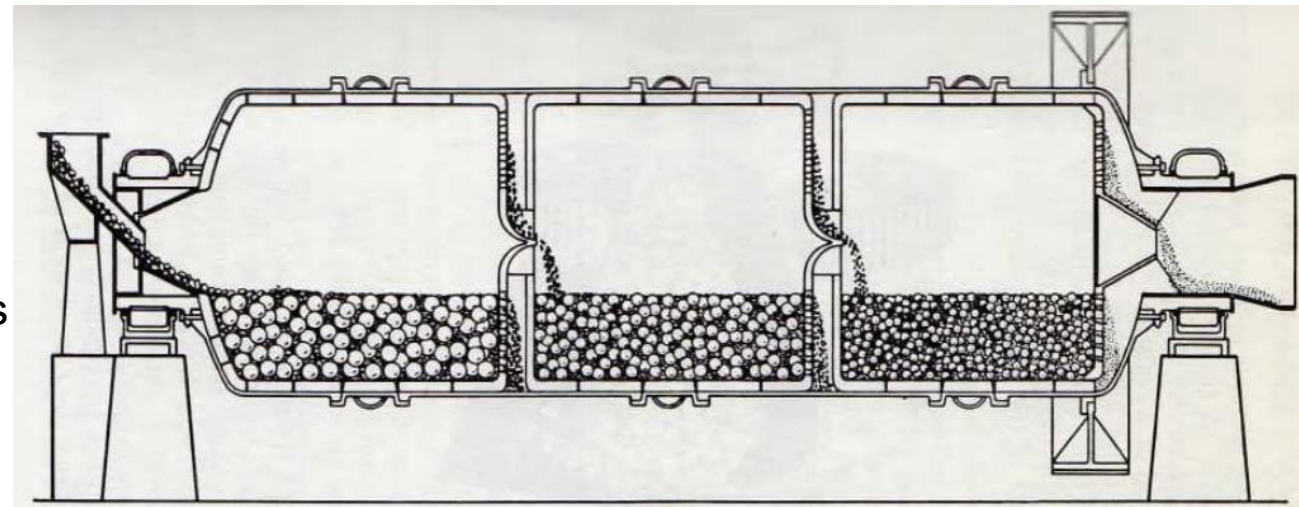
- The **large balls** are good at breaking **coarse** particles but waste energy on fine particles. The **small balls** are good at breaking **fine particles** but cannot break coarse ones. When mixed, the large balls 'interfere' with the fine grinding, and the small balls are ineffective on coarse feed.
- Compartment Mills: Slotted Partitions in tube mill
- Compartments prevent grinding media from moving between compartments.

Typical Compartment Arrangement:

- One compartment: large balls
- Second compartment: small balls
- Third compartment: pebbles
- **Large balls** only see coarse feed in the first compartment
- **Small balls** only see fine material in later compartments
- **No wasted work** – each ball size operates at its optimum particle size range

Advantage:

- Segregation of grinding media into different sizes and weights avoids wasted work
- Large, heavy balls break only large particles without interference from fines



Tumbling Mills: Conical Ball Mill

- Conical Ball Mill: Efficient Energy Use

Construction:

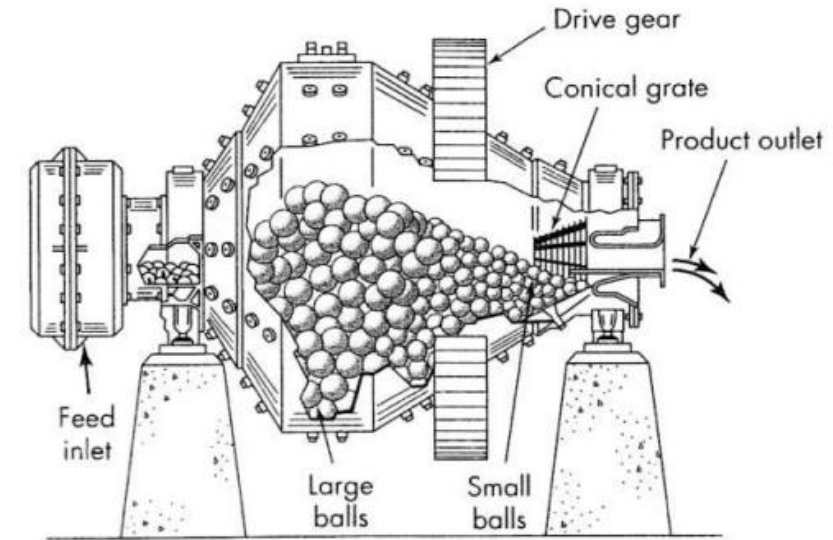
- Feed enters through a 60° cone into the primary grinding zone (maximum diameter)
- Product leaves through a 30° cone

Self-Segregation Mechanism:

- As shell rotates, large balls move toward point of maximum diameter
- Small balls migrate toward the discharge end

Efficiency Benefit:

- Initial breaking of feed particles done by largest balls dropping greatest distance
- Small particles ground by small balls dropping much smaller distance
- Energy expended suited to difficulty of breaking operation – increases mill efficiency



Tumbling Mills: Conical Ball Mill

Ball Loading Characteristics

At Rest (Mill Stopped):

- Balls occupy approximately one-half the volume of the mill
- Void fraction in the mass of balls is typically 0.40

Grinding Medium:

- May be done with dry solids
- More commonly: feed is a suspension of particles in water
- Wet grinding increases both capacity and efficiency

Mill Rotation Mechanics

Ball Movement Cycle:

- Balls picked up by mill wall
- Carried nearly to the top
- Break contact with wall and fall to bottom
- Process repeats

During Upward Movement:

- Centrifugal force keeps balls in contact with wall and with one another
- Some grinding occurs by slipping and rolling (attrition)

Primary Grinding Zone:

- Zone of impact where free-falling balls strike the bottom of the mill
- Most grinding occurs here

Tumbling Mills: Conical Ball Mill

Critical Speed of ball mill: Critical speed is the rotational speed at which the centrifugal force exactly balances the gravitational force. At this speed, the balls are pressed against the mill wall and carried all the way around without falling. The mill is said to be **centrifuging**.

When a mill is centrifuging, no grinding occurs. The balls simply rotate with the shell, stuck to the wall like clothes in a washing machine on spin cycle. Particles are not crushed because there is no impact and no relative motion between balls

Definition:

- Speed at which balls are carried over and the mill centrifuges
- No grinding occurs at or above critical speed

Formula:

- $$n_c = \frac{1}{2\pi} \sqrt{\frac{g}{R-r}}$$

Operating Rule: Operating speed n must be less than n_c

- n_c = critical speed
- g = acceleration of gravity
- R = radius of the mill
- r = radius of the grinding elements

Case Study: Paint Manufacturing Using a Ball Mill

- A paint manufacturing plant produces **decorative acrylic paints** for residential and industrial applications.
- The formulation contains **titanium dioxide (TiO_2)**, **calcium carbonate**, **color pigments**, **polymer binder**, **additives**, and **water/solvent**.
- Initially, pigments and fillers are mixed to form a slurry with pigment agglomerates ranging **100–500 μm** .
- The slurry is fed into a **ball mill** partially filled with **steel or ceramic balls**.
- As the mill rotates, the balls are lifted and then fall, producing **impact and attrition forces** that break pigment agglomerates into finer particles.
- Milling continues until the pigment particle size is reduced to approximately **5–20 μm** , ensuring good dispersion and a smooth finish.
- During production, engineers observed poor paint gloss and color uniformity due to **insufficient milling time**, leaving some pigment agglomerates unbroken.
- Increasing the milling time improved dispersion but also increased **energy consumption** and slightly raised the paint temperature, affecting viscosity.
- Engineers optimized the **ball size, ball loading, mill speed, and milling time** to achieve:
 - Uniform pigment dispersion.
 - Higher paint gloss and color consistency.
 - Improved hiding power (opacity).
 - Lower energy consumption and shorter processing time.

Ultrafine Grinders

Definition

- Equipment used to produce powders with particle sizes in the range of **1–20 μm**
- Most particles pass through a **325-mesh screen** (opening $\approx$ **44 μm**)

Applications

- Pigments and paints
- Pharmaceuticals
- Ceramics
- Food powders
- Specialty chemicals

Key Characteristics

- ✓ Very fine particle production
- ✓ Narrow particle-size distribution
- ✓ Often combined with classification system
- ✓ High energy consumption compared to coarse crushing

Types of Ultrafine Grinders

Dry Grinding

Classifying Hammer Mill

Fluid-Energy (Jet) Mill

High-Speed Hammer Mill

Wet Grinding

Agitated Mill

Stirred Media Mill

Bead Mill

Classifying Hammer Mill

Components: Feed inlet, Rotor shaft, Swing hammers, Rotor discs, Classification vanes, Air fans, Grinding chamber, product outlet and collector

Special Feature

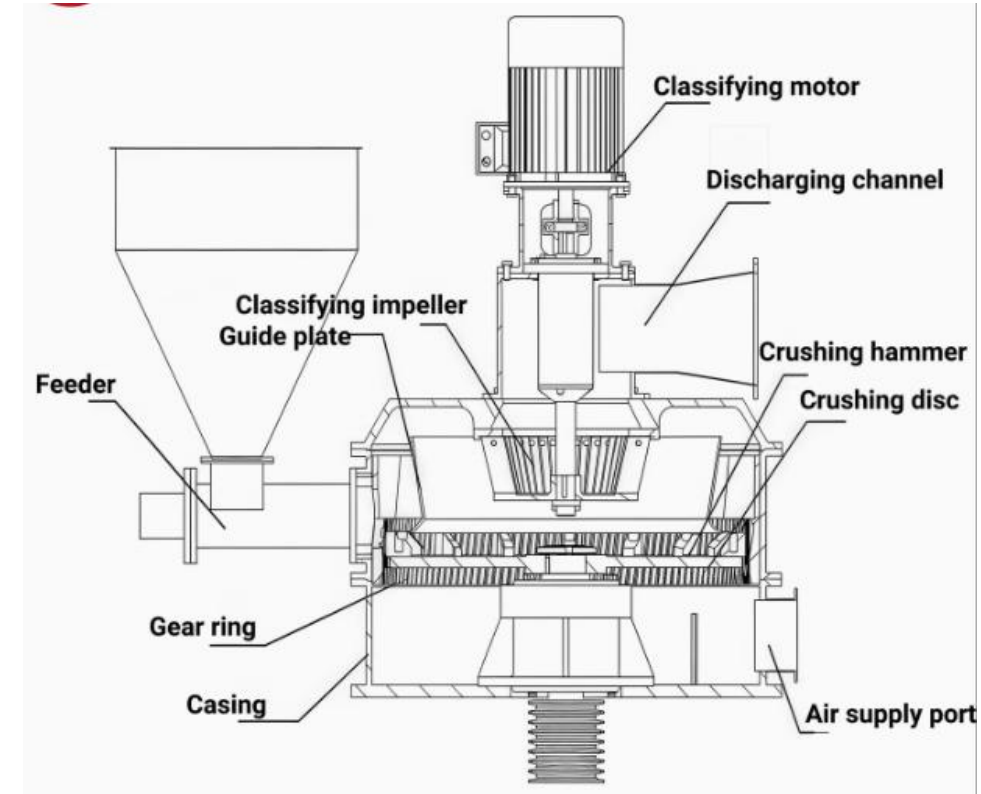
- Unlike a conventional hammer mill, this mill contains:
- Internal air classifier
- Rotor-mounted separator vanes
- Air circulation system

Function of Air Flow

- Fans create inward airflow
- Fine particles are transported out
- Coarse particles remain for further grinding

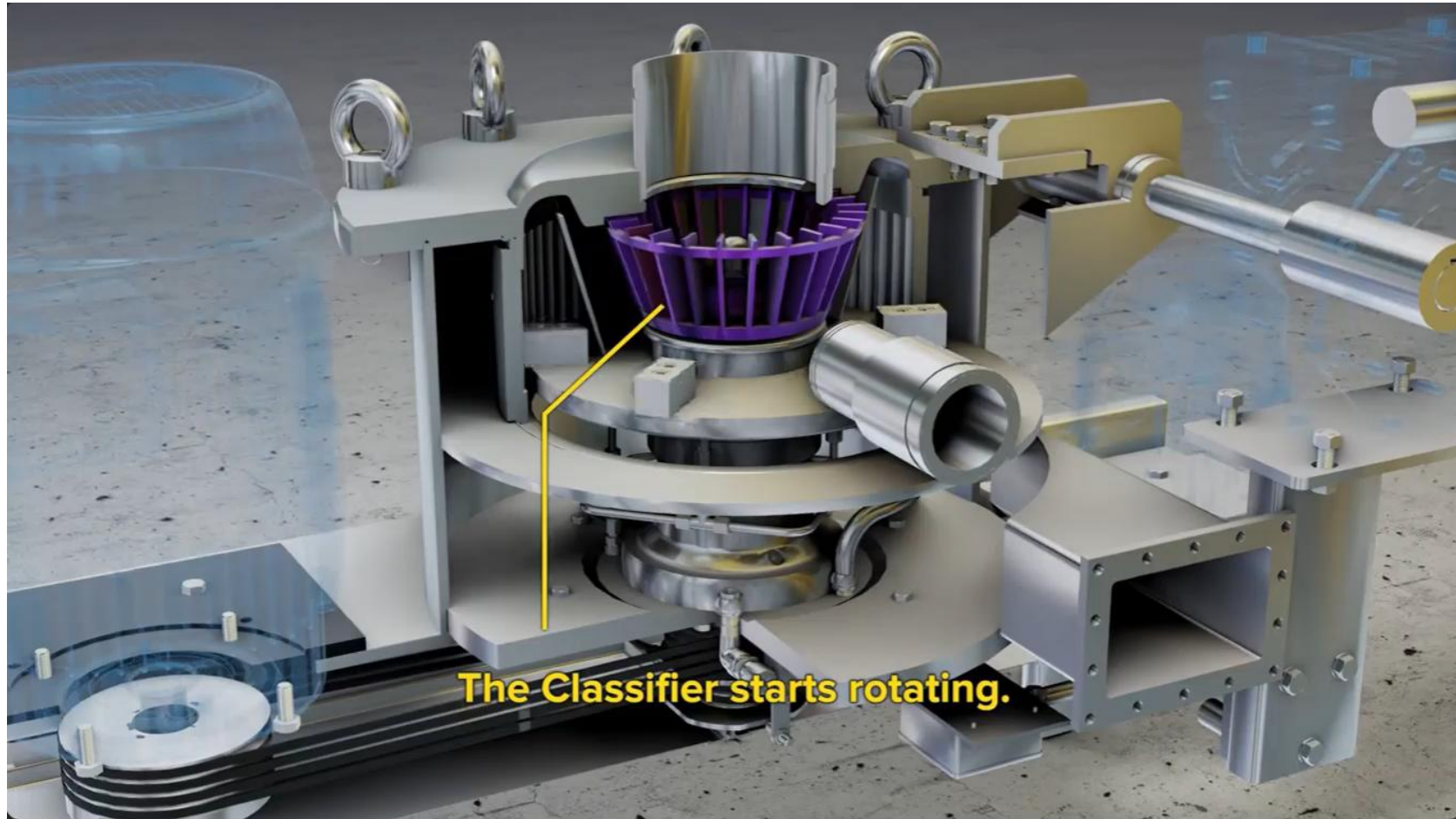
Step-by-step process:

- **Feed Entry:** Coarse material enters the grinding chamber.
- **Impact Reduction:** Swing hammers impact and break particles.
- **Air Flow:** Fans draw air **inward** toward the drive shaft.
- **Classification (Radial Vanes):**
 - **Fine particles:** Carried by air past the radial vanes → Discharge duct → Product collector.
 - **Oversize particles:** Thrown back (centrifugal force) into grinding chamber for further reduction.
- **Closed-loop:** No oversize escapes until reduced to target size.



Size reduction

Classifying Hammer Mill



Fluid Energy Mills

Definition: Mills where particles are suspended in a **high-velocity gas stream** (air or steam) and reduce size primarily by **interparticle attrition**.

Key Concept: Most of the reduction is caused by interparticle attrition.

- Particles collide with **each other**, not primarily with walls.
- **Internal classification** retains larger particles until sufficiently reduced.

Parameter	Value
Suspending gas	Compressed air or superheated steam
Gas pressure	7 atm (100 psi)
Feed particle size (max)	12 mm (½ inch)
Recommended feed size	< 100 mesh (for best efficiency)
Product size	Average 1–10 μm
Capacity	Up to 1 ton/h (loop mill up to 6,000 kg/h)

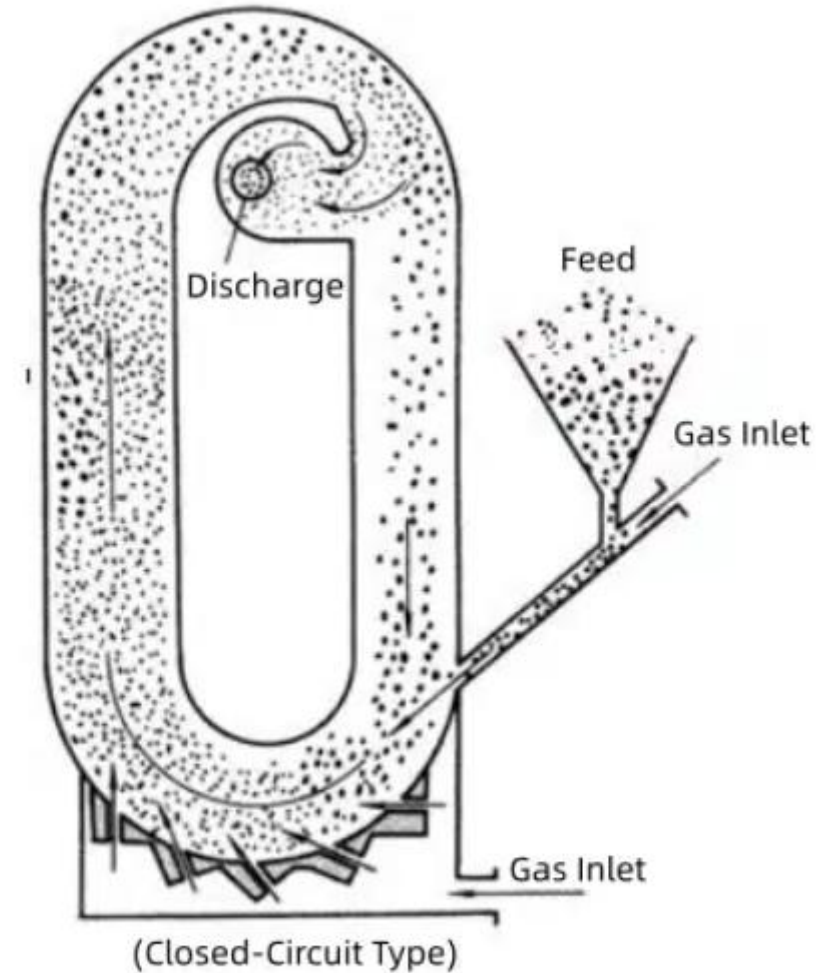
Fluid Energy Mills

Why superheated steam?

- Prevents condensation
- Higher energy content than air
- Commonly used when steam is already available onsite

Construction (Loop Mill) Components:

- **Grinding chamber:** Oval loop of pipe
 - Pipe diameter: **25–200 mm** (1–8 inches)
 - Height: **1.2–2.4 m** (4–8 feet)
- **Energizing nozzles:** Inject air/steam at 7 atm
- **Venturi injector:** Feeds solids at bottom
- **Classifier outlet:** At upper bend of loop
- **Discharge:** Leads to cyclone + bag collector



Fluid Energy Mills

Working Principle

- **Feed entry:** Solid particles enter near the **bottom** through a **venturi injector** (creates suction).
- **Acceleration:** High-velocity gas (7 atm) from **energizing nozzles** accelerates particles to near-sonic speeds.
- **Attrition:** Particles collide with each other in turbulent flow. Dominant mechanism: **particle-on-particle impact**.
- **Classification at upper bend:**
 - As gas flows around the **curved bend** at high speed:
 - **Coarse particles** → thrown **outward** against outer wall (centrifugal force)
 - **Fine particles** → congregate at **inner wall**
 - **Swirl patterns** aid separation.
- **Product removal:** Discharge opening at **inner wall** → cyclone separator → bag collector.
- **Recirculation:** Oversize particles continue circulating in the loop.

	Force	Effect on Coarse Particles	Effect on Fine Particles
Why does classification happen at the upper bend?	Centrifugal	High → thrown outward	Low → stay inward
	Drag (gas flow)	Moderate	High relative to mass

Agitated Mills (Wet Grinding)

Small, non-rotary mills containing a **solid grinding medium** suspended in liquid.

Grinding media examples:

- Steel balls
- Ceramic pellets
- Sand grains

Typical applications:

- Silica
- Titanium dioxide (TiO₂) – paint pigment
- Pharmaceuticals (wet milling)

Construction & Operation

Vessel: Vertical cylinder, Capacity: **4 to 1,200**

L (1 to 300 gallons), Filled with liquid + grinding media

Design

Impeller type

Vibratory type

Agitation Method

Multi-armed impeller stirs the charge

Reciprocating central column at **~20 Hz** (vibration)

Best For

General ultrafine grinding

Hard materials (silica, TiO₂)

Flow pattern:

- **Feed:** Concentrated slurry at **top**
- **Product withdrawal:** Through **screen at bottom** (retains grinding media)
- **Product size:** **1 µm or finer**

Agitated Mills (Wet Grinding)

Working Principle

- **Step-by-step:**
- **Fill vessel** with grinding media (balls/sand) + liquid.
- **Add feed slurry** at top (concentrated suspension of solid).
- **Agitate** via impeller or vibration:
 - Media moves violently
 - Particles get trapped between media
 - Size reduction by **compression, impact, and shear**
- **Internal classification:** Screen at bottom allows only fine product + liquid to exit. Grinding media remains inside.
- **Continuous or batch operation** possible.
- **Key advantage:** Can reach **sub-micron** particles (below 1 μm).

Typical applications:

Industry	Material	Target Size
Paints & coatings	Titanium dioxide (TiO_2)	< 1 μm (color uniformity)
Electronics	Silica for polishing slurries	0.5 μm
Pharma	Drug nanoparticles	200–500 nm
Minerals	Kaolin clay	< 2 μm (paper coating)

Fluid Energy Mills vs. Agitated Mills

Feature	Fluid Energy Mill	Agitated Mill
Grinding environment	Dry (gas)	Wet (liquid slurry)
Grinding mechanism	Interparticle attrition	Media impact + shear
Moving parts	None (gas-driven)	Impeller or vibrating column
Feed size	Up to 12 mm	Typically pre-ground
Product size	1–10 μm	$\leq 1 \mu\text{m}$ (sub-micron)
Capacity	Up to 6,000 kg/h	Batch: 4–1,200 L
Utilities	Compressed air/steam at 7 atm	Electricity + cooling water
Best for	Dry, non-sticky powders	Hard materials, slurries

Selection guide: Dry powder, 1–20 μm → Classifying hammer mill; ; Dry powder, 1–10 μm , heat-sensitive → Fluid energy mill;; Wet slurry, < 1 μm → Agitated mill

Colloid Mills

Core Principle: *Intense Fluid Shear in a High-Velocity Stream.*

- **Primary Goal:** Not to "crush," but to **disrupt lightly bonded clusters** (agglomerates) or disperse droplets.
- **Mechanism:** Creates a stable **suspension** (solid-in-liquid) or **emulsion** (liquid-in-liquid).
- **Final Size:** Particles/Droplets $< 5\ \mu\text{m}$ (Sub-micron possible).

Applications & Stabilization

Industry

Food

Pharma

Paints/Coatings

Lubricants

Product

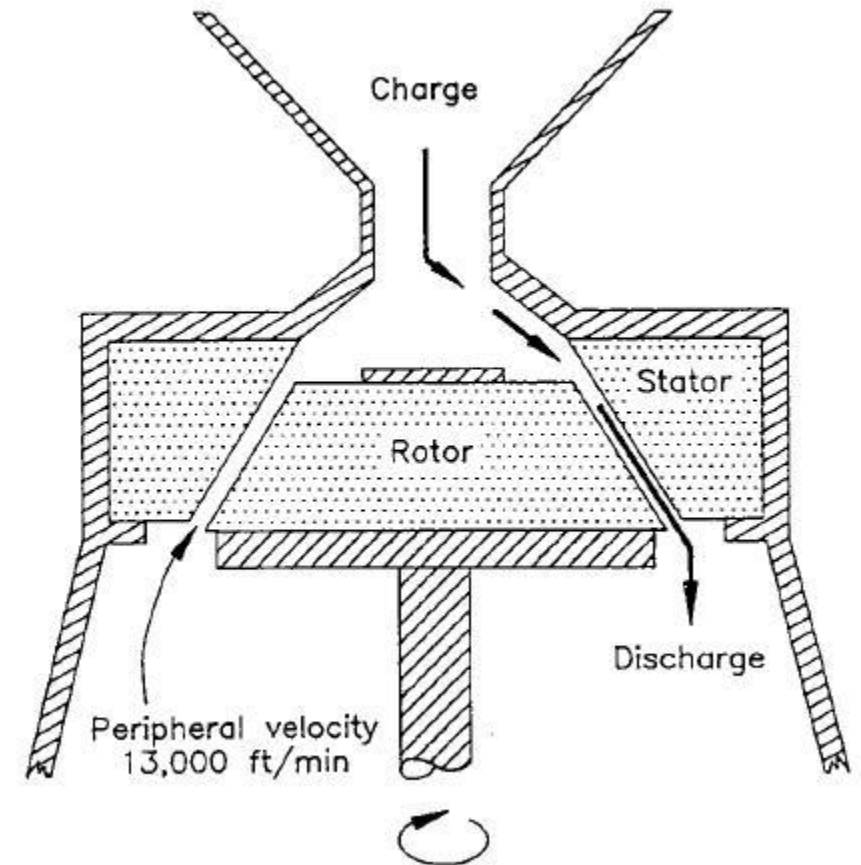
Syrups, Purees, Milk

Ointments

Pigment dispersions

Greases

Stabilization Requirement: Chemical additives (surfactants, emulsifiers) are **often required** to prevent re-agglomeration/re-coalescence



Colloid Mills

Working Principle & Geometry

- **Key Component:** Closely spaced surfaces (one stationary, one rotating).
- **Rotor speed:** Peripheral velocity ≥ 50 m/s.
- **Design example:** Disk-shaped rotor rotating inside a casing.
- **Critical Gap (Clearance):** Adjustable down to **25 μm** (human hair thickness).

Heat Generation & Cooling: Intense shear produces significant frictional heat. **Cooling is often required** (water jacket around stator).

Transition – When Colloid Mills Fail?

Some materials are: **Too tenacious** (tough, difficult to break), **Too resilient** (elastic, springs back)

Alternative Approach: Cutting Machines

Colloid Mills

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Enlargement

The Problem with Fine Particles

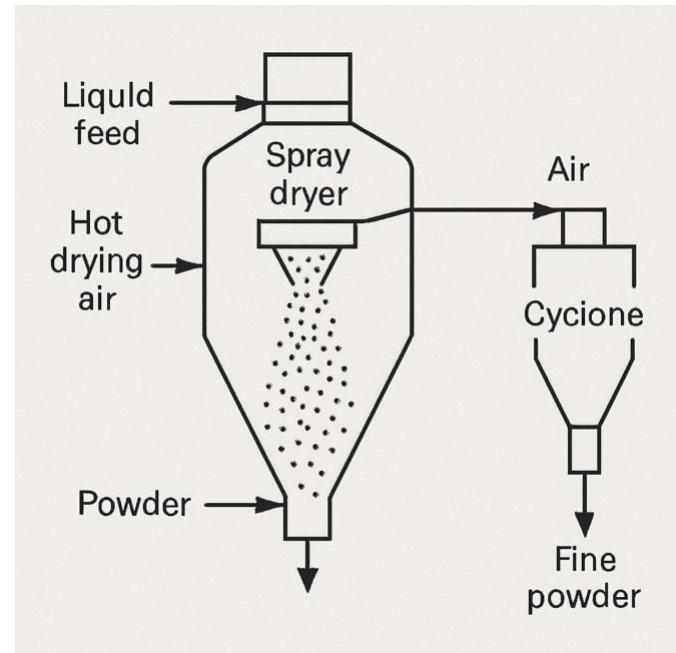
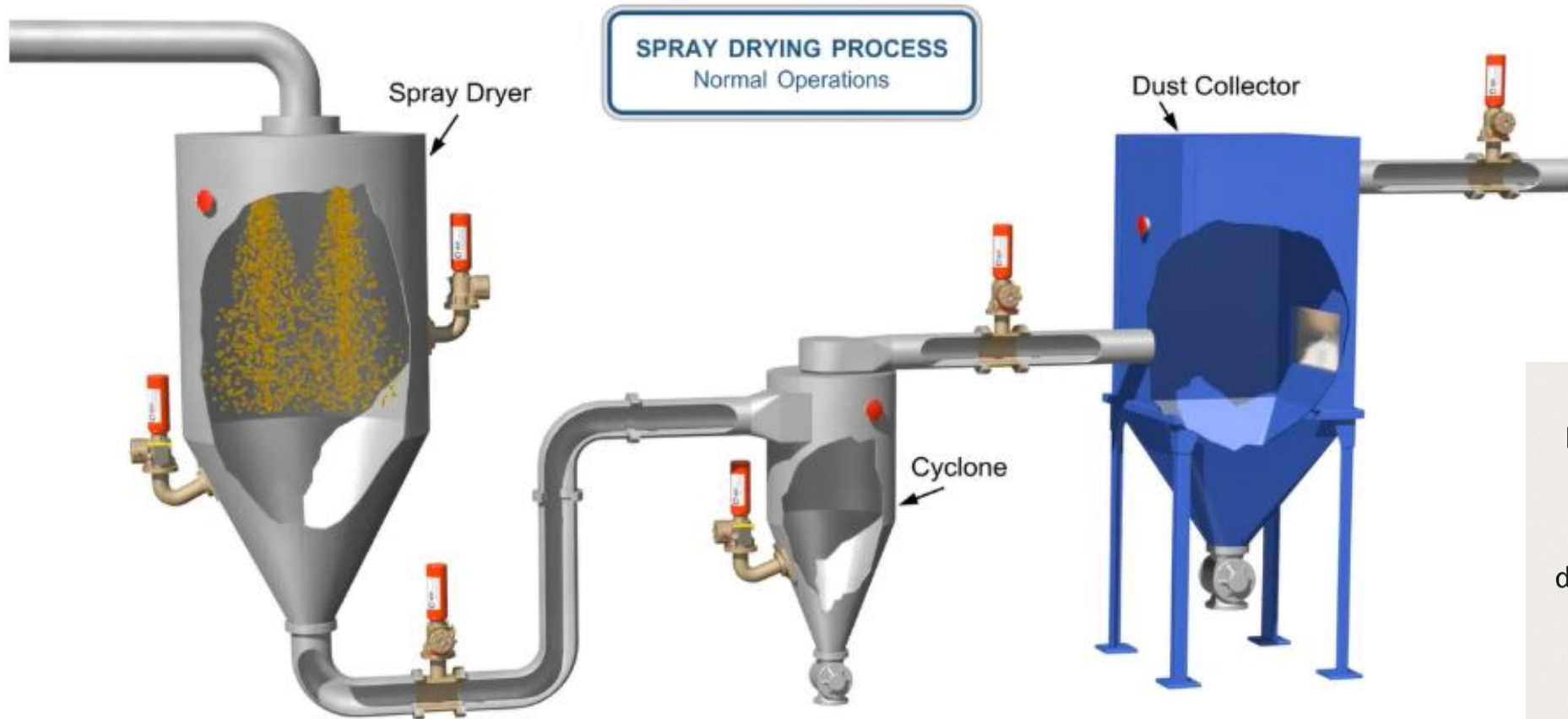
SIZE ENLARGEMENT

- 1. Poor Flowability:** Bulk powders do not flow smoothly. Fine particles cling to each other, making handling difficult.
- 2. Fluidization Failure:** In gas-solid systems, fine particles are difficult to fluidize. Instead of even dispersion, channelling occurs — gas bypasses through preferred paths without contacting particles properly.
- 3. Hopper Discharge Issues:** Fine particles cling to hopper walls and form stable bridges or arches at the discharge point, blocking flow completely.
- 4. Vibration & Stirring Are Not Enough:** Although vibration or mechanical stirring may minimize the problems, they cannot overcome them entirely. A permanent solution is needed.
- 5. Dust Formation:** During loading into vehicles, fine particles form dust clouds, creating safety and environmental hazards.
- 6. Wind Dispersal:** In windy conditions, fine particles become dispersed over long distances, causing widespread contamination even if the material is chemically inert.
- 7. Respiratory Health Hazards:** Particle size is a critical factor for health. Very fine particles may be exhaled, while those in the respirable range deposit in the lungs. Even inert materials become hazardous when inhaled.
Example – Asbestos: The serious health hazard of asbestos is largely associated with its particle size range, particle shape, and its tendency to collect in the lungs.

Size Enlargement: Spay Drying

- 1. What is Spray Drying?** Spray drying is a size enlargement process where a liquid or suspension is atomized into fine droplets. The solvent is rapidly evaporated, leaving behind dried solid aggregates.
- 2. How Particle Size is Determined:** The size of the final dried particles is largely determined by the size of the droplet of liquid or suspension produced during atomization. Droplet size can be controlled by using a suitably designed spray nozzle.
- 3. The Binding Mechanism:** The aggregates of dried material are held together as a result of the deposition of small amounts of solute on the surface of the particles during drying. This solute acts as a natural binder, giving mechanical strength to the final granules.
- 4. Factors Affecting Droplet Size:** For a given nozzle design, the droplet size depends on several operating parameters. Flow rate of the liquid through the nozzle directly influences droplet size. Liquid properties affect droplet size significantly, with viscosity being the most important property. Outlet temperature has an effect on droplet size, but to a lesser extent compared to flow rate and viscosity.
- 5. Effect of Viscosity:** Viscous liquids tend to form large drops. Large drops yield large aggregates after drying. Low viscosity leads to finer droplets and smaller dried particles. High viscosity leads to larger droplets and larger dried particles.

Spray Drier



Spay Drier

Step 1: Prepare the liquid feed (solution, suspension, emulsion, slurry, or gel). Pump the feed to an atomizer located at the top of the drying chamber.

The atomizer disintegrates the bulk liquid into fine droplets using pressure energy, centrifugal kinetic energy (spinning discs/cups), or gaseous energy (twin-fluid).

This creates an extremely large interfacial area for rapid evaporation.

Step 2: Liquid exits the atomizer as thin sheets or threads which break into droplets due to instability (Rayleigh theory: wavelength $\approx 4.5 \times$ jet diameter).

The fine droplets are sprayed into the drying chamber where they contact a current of hot gas (co-current or counter-current flow).

Step 3: Due to the large interfacial area, solvent evaporates at a very high rate.

The droplet temperature remains below the wet bulb temperature of the drying gas until drying is nearly complete.

This low temperature protects heat-sensitive materials such as milk, coffee, and plasma from deterioration.

Step 4: As solvent evaporates, solute deposits on particle surfaces act as natural binders, forming solid aggregates.

The dried powder is separated from the exhaust gas using cyclones or bag filters and collected as the final product in a finely divided state.

Exhaust gas is then treated or released.

Size Enlargement: Spray Drying

1. **Basic Principle:** Water is evaporated from a solution or suspension of solid particles by spraying the mixture into a vessel through which a current of hot gases is passed.
2. **Atomizer:** An atomizer is a device which: Causes liquid to be disintegrated into drops lying within a specified size range and Controls the spatial distribution of those drops.
3. **Advantage:** A large interfacial area is produced by atomization. This results in a high rate of evaporation.
4. **Temperature Control:** Drop temperatures remain below the wet bulb temperature of the drying gas until drying is almost complete.
5. **Heat-Sensitive Products:** substances successfully dried by spray drying include: Milk, Coffee, Plasma
6. **Final Product Characteristics:** Because of the fine state of subdivision of the liquid during atomization, the dried material is obtained in a finely divided state.
7. **Types of Liquids That Can Be Processed:** Solutions, Emulsions, Dispersions, Slurries, Gels
8. **Challenge:** When atomizers are used for slurries, pastes, and liquids with anomalous properties:
There is a great deterioration in performance: Atomizers may be rapidly eroded and damaged; They may become completely useless

Size Enlargement: Prilling Tower

What is Prilling?

- Prilling is a size enlargement technique in which:
- Molten material is converted into droplets.
- Droplets are introduced at the top of a tall tower.
- Droplets fall downward against an upward stream of air.
- During falling, droplets cool and solidify into spherical particles called **prills**.

Applications: Urea, Ammonium nitrate, Fertilizers, Waxes

Slide 3: Working Principle of Prilling

1. Melt preparation
2. Droplet formation through nozzles
3. Droplets released from tower top
4. Upward air flow cools droplets
5. Solid prills collected at bottom

Key Features

- Tall and narrow tower
- Counter-current air flow
- Produces nearly spherical particles
- Larger particles than spray drying

Advantages

- ✓ Uniform particle size
- ✓ Good flowability
- ✓ Simple operation

Limitations

- X Requires tall towers
- X Limited to meltable material

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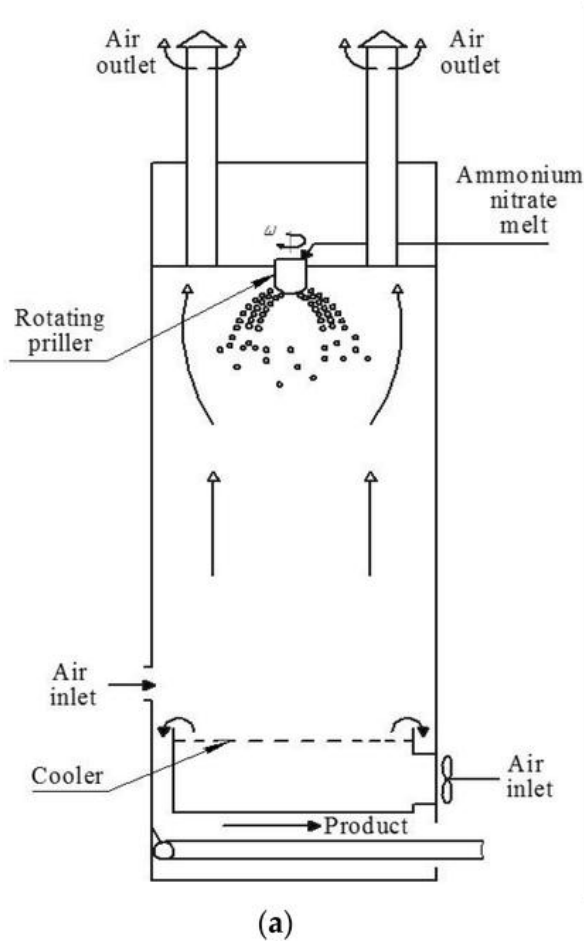
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- ✓ Simple operation

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- X Limited to meltable material

Enlargement

Size Enlargement: Prilling Tower



Working Process of a Prilling Tower

A **prilling tower** is used to convert a molten material into solid, spherical particles called **prills**.

Step 1: Feed Introduction: The material (e.g., molten urea or ammonium nitrate) is heated above its melting point. The molten liquid is pumped to the top of the prilling tower.

Step 2: Droplet Formation: At the top of the tower, the molten liquid passes through a rotating bucket, perforated plate, or spray nozzle. The liquid breaks into uniform droplets due to centrifugal force or gravity.

Step 3: Free Fall Through the Tower: The droplets fall downward through the tall tower under gravity. An upward flow of cool air is introduced from the bottom of the tower (counter-current operation)

Step 4: Cooling and Solidification: Heat is removed from the falling droplets by the air stream. The droplets gradually cool and solidify while maintaining a nearly spherical shape due to surface tension.

Step 5: Collection of Prills: By the time the particles reach the bottom of the tower, they have completely solidified. The solid spherical particles (prills) are collected and conveyed for storage, screening, or packaging.

Size Enlargement: Fluidized Bed Agglomeration

Fluidized Bed Agglomeration: Fluidized bed agglomeration is a size enlargement process in which a liquid binder is sprayed onto a bed of fluidized particles, causing them to stick together and form larger granules.

Working Principle

- Hot air enters from the bottom and fluidizes the particles.
- A binder solution or suspension is atomized through a spray nozzle.
- Wet particles collide and adhere to each other.
- Continuous drying by hot air solidifies the granules.
- Repeated deposition of solids increases particle size.

Particle Growth Mechanism

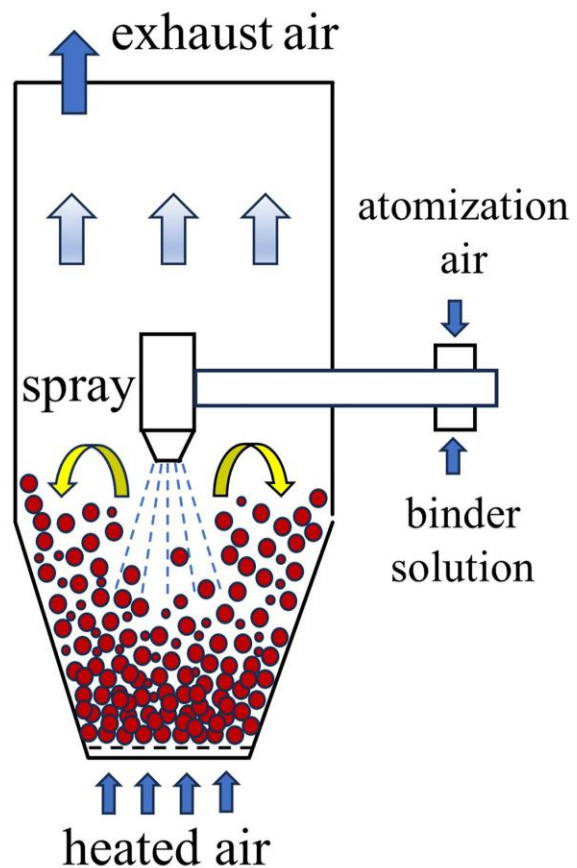
1. Wetting of particle surface
2. Particle-particle collision
3. Formation of liquid bridges
4. Drying and solidification
5. Growth of granules by successive layer deposition

Advantages

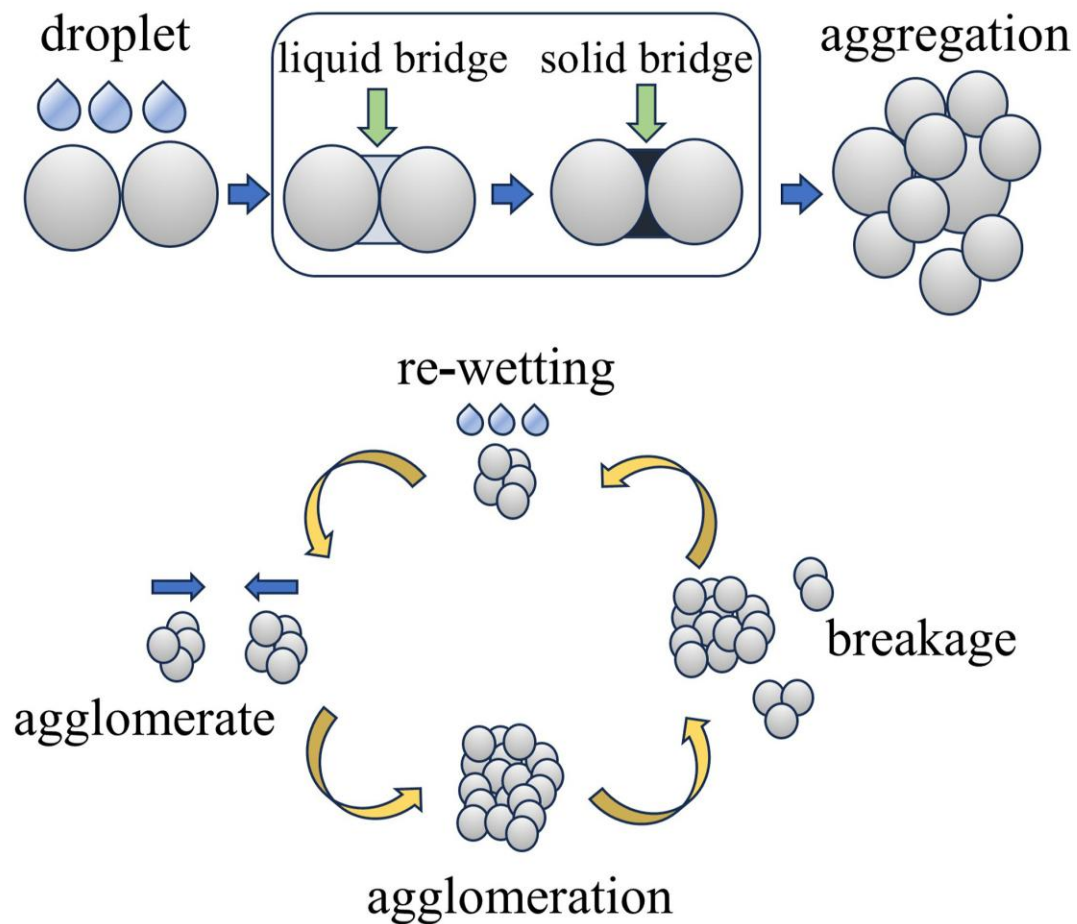
- ✓ Simultaneous agglomeration and drying
- ✓ Uniform particle growth
- ✓ Good heat and mass transfer
- ✓ Improved flowability and handling
- ✓ Continuous operation possible

Size Enlargement: Fluidized Bed Agglomeration

Fluidized bed granulation



Droplet-particle interaction



Size Enlargement: Fluidized Bed Agglomeration

Fluidized Bed Agglomeration: Process Steps

- **Fluidization** – Hot air suspends and mixes particles in the bed.
- **Binder Spraying** – Atomized binder droplets are sprayed onto fluidized particles.
- **Particle Wetting** – Droplets wet particle surfaces, making them sticky.
- **Liquid Bridge Formation** – Colliding wet particles form temporary liquid bridges.
- **Drying & Solid Bridge Formation** – Hot air evaporates the liquid, creating permanent solid bonds.
- **Agglomerate Growth** – Repeated wetting and bonding increase granule size.
- **Breakage & Re-agglomeration** – Some granules break due to collisions and rejoin during further spraying.
- **Final Granules** – Stable enlarged particles are obtained when growth and breakage reach equilibrium.

Industrial Applications: Fertilizers, Pharmaceutical granules, instant food powders, Detergents, Catalyst particles

Size Enlargement: Drum Agglomerator

- **Core Principle:** Particles are tumbled inside an open, rotating cylinder with roughened interior walls. They experience a combination of gravitational forces (rolling down) and centrifugal forces (being lifted up).
- **Key Mechanism:** As the drum rotates, particles stick to the roughened wall, are lifted to a certain height, and then cascade down. With each tumbling cycle, fines collide and stick together, forming larger agglomerates.
- **Role of Liquid:** Liquid (binder solution or water) is sprayed onto the bed surface or introduced through distribution pipes under the bed to promote particle adhesion.
- **Main Components: Rotating Cylindrical Shell:** Horizontally or slightly inclined open drum, often made of steel with a roughened internal surface (rubber lining or welded bars)
- **Drive System:** Motor and gearbox with trunnion rollers or tire rings that rotate the drum at controlled speed
- **Liquid Spray System:** Nozzles mounted above the bed or perforated pipes placed under the bed for uniform moisture addition
- **Feed Chute:** For continuous introduction of fine powder at one end
- **Discharge Hood:** At the opposite end where enlarged product exits

Size Enlargement: Drum Agglomerator

Working Mechanism :

Step 1 – Feeding: Fine powder enters continuously from the inlet end.

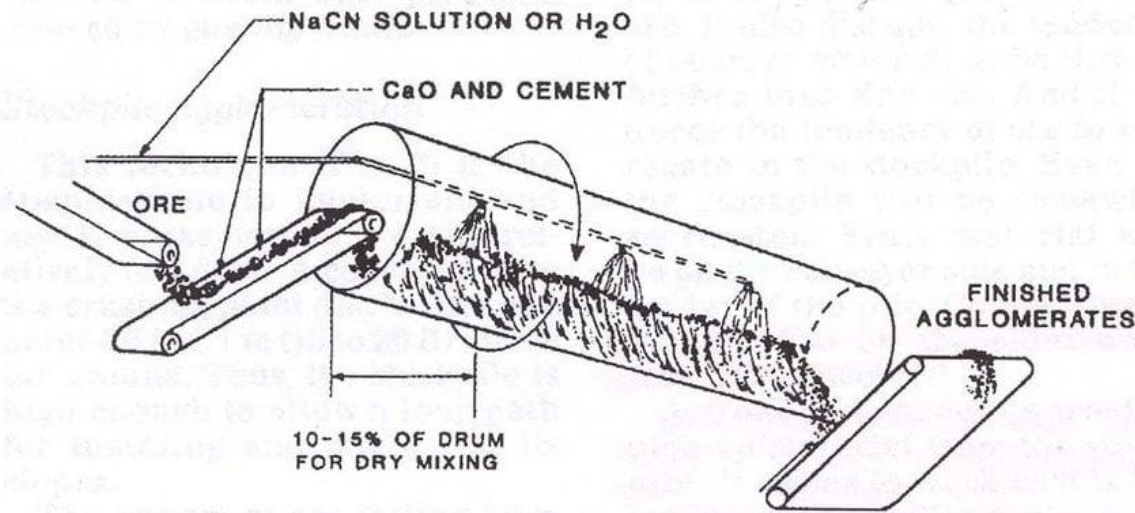
Step 2 – Liquid Addition: Binder liquid is sprayed onto the moving bed to wet particle surfaces.

Step 3 – Tumbling & Nucleation: As the drum rotates, particles are lifted by the roughened wall and then cascade down. Small nuclei granules form from wetted fines.

Step 4 – Growth & Consolidation: With continued tumbling, nuclei roll over the bed, picking up more fines and compacting into larger, rounded agglomerates. Mean retention time: 60 to 120 seconds.

Step 5 – Discharge: Finished agglomerates exit from the opposite end as the drum rotates.

Similar Equipment: A paddle mixer achieves a similar action but with centrifugal forces dominating over gravity.



Drum Agglomerators are used for: Iron ore pelletization, Fertilizer granulation, Cement clinker, Animal feed, Fly ash agglomeration

Size Enlargement: Pan Agglomerator

Core Principle:

A rotating, shallow pan (disc) inclined at an angle processes particles similarly to a drum, but with an added classifying action. Due to the pan's geometry and rotation, fines preferentially stay longer near the bottom of the pan, while larger, denser agglomerates migrate to the rim and overflow or are discharged.

Main Components:

Inclined Rotating Pan (Disc): Shallow, bowl-shaped vessel mounted on a central shaft, with an adjustable inclination angle (typically 40° to 65°)

Drive Motor & Gearbox: Provides controlled rotation speed for the pan

Fixed Scraper Blade: Stationary blade positioned inside the rotating pan to prevent buildup and assist in mixing

Liquid Spray System: Nozzles positioned above the pan for uniform wetting

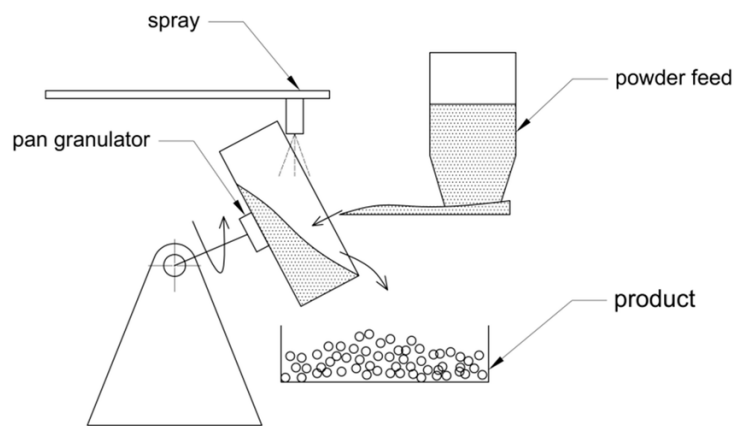
Feed System: Conveyor or chute delivering powder near the bottom of the pan

Discharge Lip/Edge: The rim of the pan where classified agglomerates spill over into a collection chute

Size Enlargement: Pan Agglomerator

Working Mechanism

1. **Feeding:** Fine powder is introduced continuously near the bottom (lowest point) of the rotating pan.
2. **Liquid Addition:** Binder solution is sprayed onto the tumbling bed as the pan rotates.
3. **Tumbling & Layering:** The inclined rotation causes particles to tumble in a rolling motion. Fines stay in the lower zone due to the classifying action.
4. **Preferential Retention:** Small or low-density particles remain longer in the pan, while larger, denser, stronger agglomerates move outward toward the rim.
5. **Size-Based Discharge:** Only agglomerates that have reached the desired size and strength overflow the pan's rim automatically. Undersized material continues circulating.
6. **Collection:** Finished product is collected from the chute below the discharge lip.



Applications: High-density iron ore concentrate, Phosphate rock, Carbon black, Ceramic powders, Recycling of steel mill dusts

Size Enlargement: Pug Mill & Extruder System

A **Pug Mill** is a continuous mixing and kneading machine used to blend solid materials with small amounts of liquid. An **Extruder** is a device that forces material through a shaped die to create a uniform cross-section.

Primary Purpose:

- To achieve **high-density compaction** of fine solids.
- To combine **mixing, kneading, and shaping** in a single operation.
- To transform a powder or crumbly feed into a cohesive, plastic mass (extrudate).

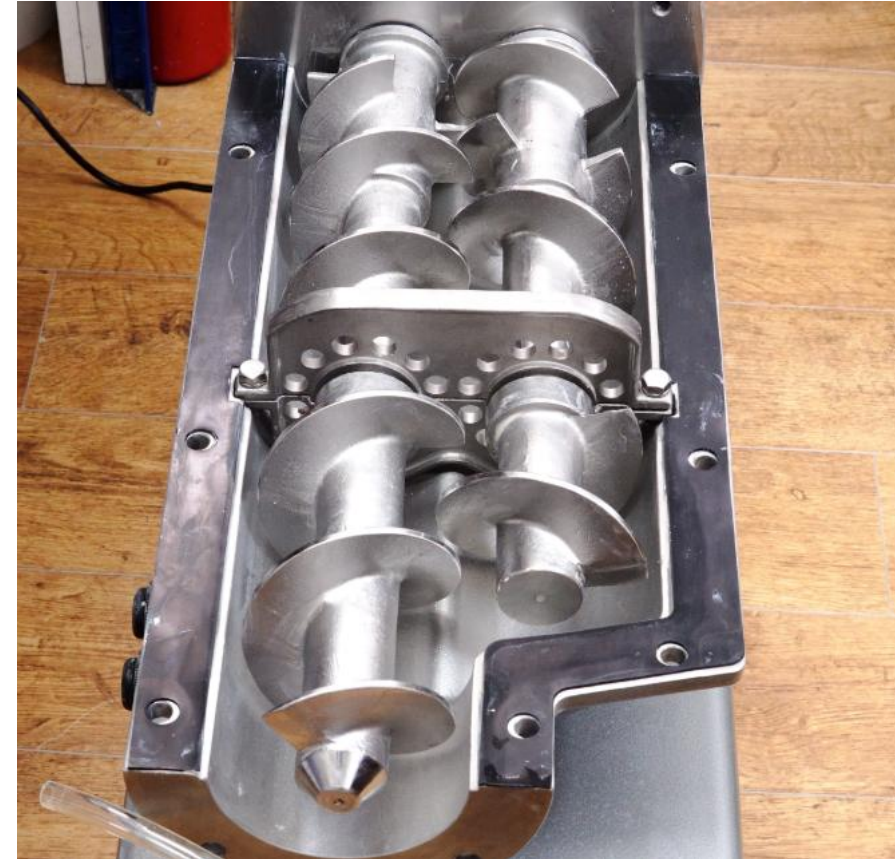
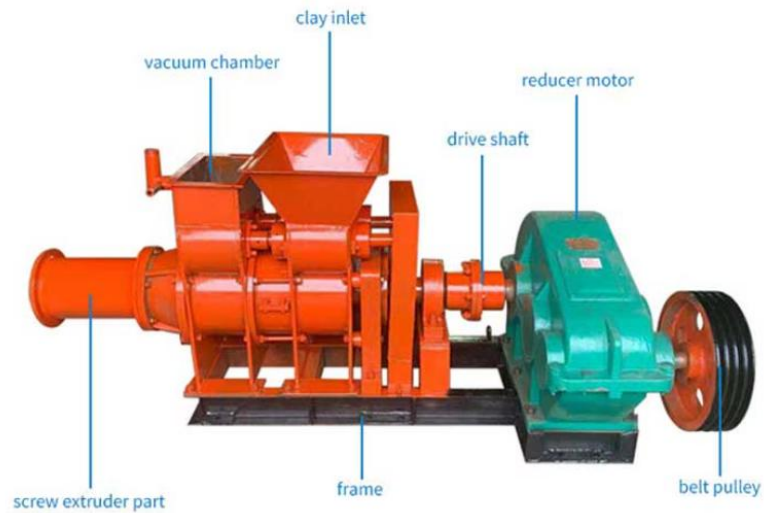
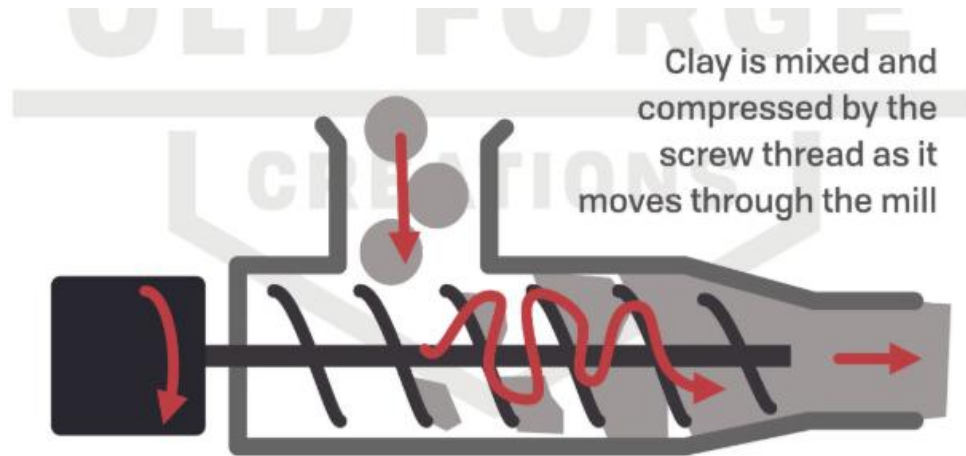
Principle:

- The machine applies **intense shear and kneading action** to a low-moisture feed.
- **Combined Forces:** The material is simultaneously pushed (ribbing), cut (shearing), and folded (kneading).
- **Energy Conversion:** High mechanical energy input is converted into heat, significantly raising the material's temperature (often 50-80°C above ambient).
- **Result:** This heat, combined with pressure, plasticizes the material, allowing densification and extrusion without adding large amounts of water.

Main Components

- **Trough / Barrel:** A heavy-duty stationary casing containing the mixing elements.
- **Twin Shafts:** Two parallel, counter-rotating shafts fitted with paddles or helical ribbons.
- **Paddles / Blades:** Special angled paddles that create the kneading and forward conveying action.
- **Die Plate:** A perforated plate at the discharge end containing shaped holes (round, star, or custom profiles).
- **Drive Motor & Gearbox:** Provides high torque at low speed to overcome the resistance of dense material.

Size Enlargement: Pug Mill & Extruder System



Size Enlargement: Pug Mill & Extruder System

Working Mechanism

1. **Feed Entry:** Damp powder (10-25% moisture) enters the trough via a hopper.
2. **Initial Mixing:** Twin counter-rotating shafts with paddles lift, fold, and shear the powder, breaking agglomerates.
3. **Kneading & Plasticization:** Intense mechanical action generates frictional heat, softening binders and coating particles with liquid film.
4. **Densification:** Air is expelled as the material is compressed into a dough-like, plastic consistency.
5. **Extrusion:** The plastic mass is forced forward by the paddles and pushed through the die plate.
6. **Product Output:** Long strings or pellets (extrudate) are cut to length by a rotating knife.

Key Factors Affecting Performance

- **Feed Moisture Content:** Too dry = no binding; too wet = sticky, soft extrudate. Optimum is usually 10-15%.
- **Residence Time:** Controlled by paddle angle and shaft speed. Longer time = more shear and densification.
- **Temperature Rise:** High energy input can degrade heat-sensitive materials. Cooling jackets on the trough are essential.
- **Die Design:** Hole diameter, length, and shape affect back-pressure and extrudate quality.

Size Enlargement: Pug Mill & Extruder System

Industrial Applications

- **Catalyst Manufacturing:** Extruding cylindrical alumina or zeolite pellets for packed bed reactors.
- **Fertilizer Production:** Densifying and granulating potash, DAP, or NPK blends to prevent dust and segregation.
- **Carbon & Charcoal:** Producing activated carbon pellets for gas adsorption filters.
- **Ceramics & Refractories:** Forming brick and tile shapes from clay bodies.
- **Animal Feed & Biomass:** Pelletizing animal feed or wood waste for biofuel briquettes.

Advantages

- **High Densification:** Achieves high bulk density from low-density dust.
- **Dust-Free Product:** Eliminates handling hazards (explosions, toxicity).
- **Single Operation:** Mixes, kneads, and extrudes in one machine (low footprint).
- **Tunable Residence Time:** Can be adjusted for specific plasticization needs.

Limitations:

- **High Energy Cost:** Significant electricity consumption and wear on paddles.
- **Heat Sensitivity:** Not suitable for materials that degrade or melt under frictional heating (e.g., certain polymers or sugars).
- **Moisture Sensitivity:** Requires precise feed moisture; too dry or too wet blocks the die.
- **High Wear:** Abrasive materials (e.g., sand, coke) rapidly erode paddles and die plates.

Mechanical Separations

Two General Methods of Mechanical Separation

Method 1: Screening / Membrane

- **Principle:** Use a screen, septum, or porous membrane.
- **Action:** Retains one component, allows the other to pass.
- **Application:** Solid-solid separation by size; also for macromolecules (proteins, polymers) in solution.

Method 2: Sedimentation

- **Principle:** Utilize differences in movement speed through a fluid.
- **Action:** Particles or drops settle at different rates (based on size, density, shape).
- **Application:** Thickening, Clarification, Cyclone separation.

Screening

1. Basic Principle

- A method of separating particles **by size alone**.
- Solids are dropped or thrown against a screening surface.
- **Fines (Undersize)** → Pass through the openings.
- **Tails (Oversize)** → Retained on the surface.

2. Single vs. Multiple Screens

One screen → One separation into **two unsized fractions**.

*Unsize*d: Only one size limit (upper or lower) is known.

Series of screens → **Sized fractions**.

Sized: Both maximum *and* minimum particle sizes are known.

3. . Mesh size is the number of openings per linear inch.

- Example: 150- mesh size= 150x150 openings per square inch
- Opening size depends on wire thickness

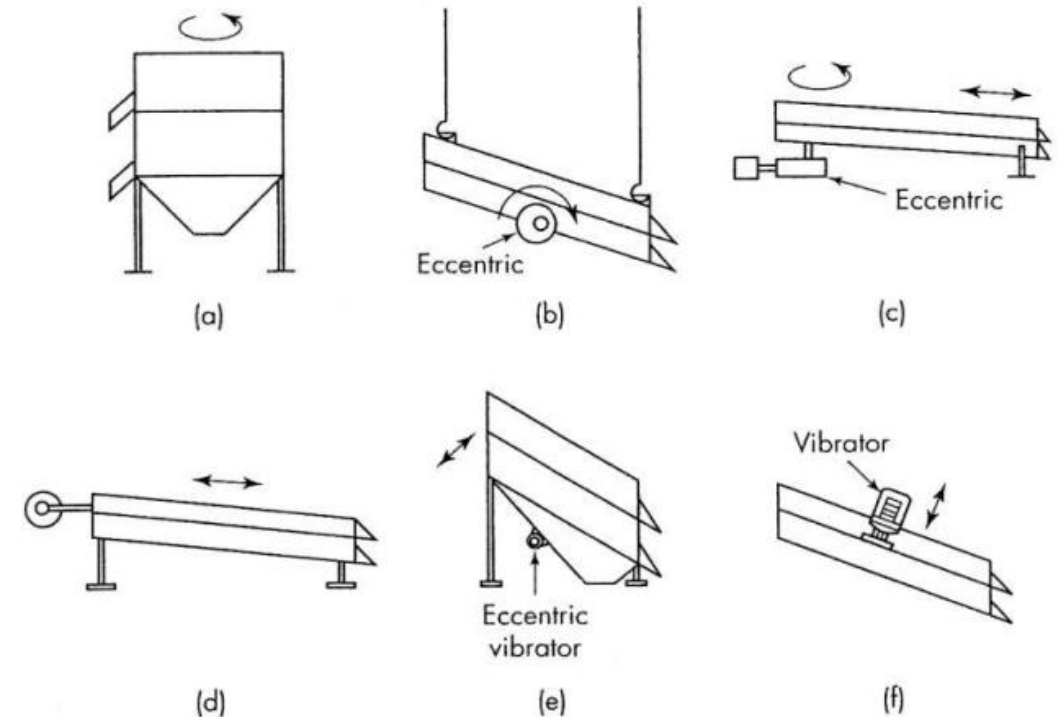


FIGURE 29.1

Motions of screens. (a) gyrations in horizontal plane; (b) gyrations in vertical plane; (c) gyrations at one end, shaking at other; (d) shaking; (e) mechanically vibrated; (f) electrically vibrated.

Stationary Screens & Grizzlies

1. What is a Grizzly?

- A **grid of parallel metal bars**.
- Mounted in an **inclined, stationary frame**.
- Slope direction is parallel to the length of the bars.

2. Typical Feed & Application

- Handles **very coarse feed** (e.g., discharge from a primary crusher).
- Feed enters at the **upper end** of the grid.

3. Separation Mechanism

- **Large chunks** → Roll and slide down to the **tails discharge** (oversize).
- **Small lumps** → Fall through bar spacing to a **separate collector** (undersize).

4. Bar Spacing Range

- **2 to 8 inches** (50 to 200 mm) between bars.
- Designed for extremely coarse separations.

5. Limitation

- Effective **only** when all three conditions are met:
- Very **coarse** solids.
- **Free-flowing** material.
- **Few fine particles** present.



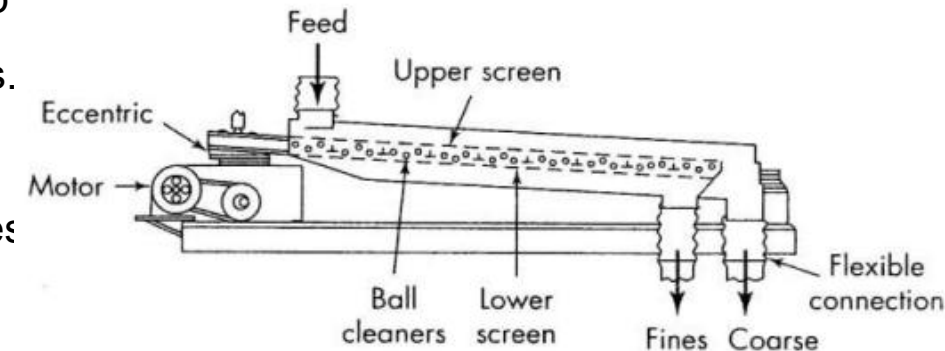
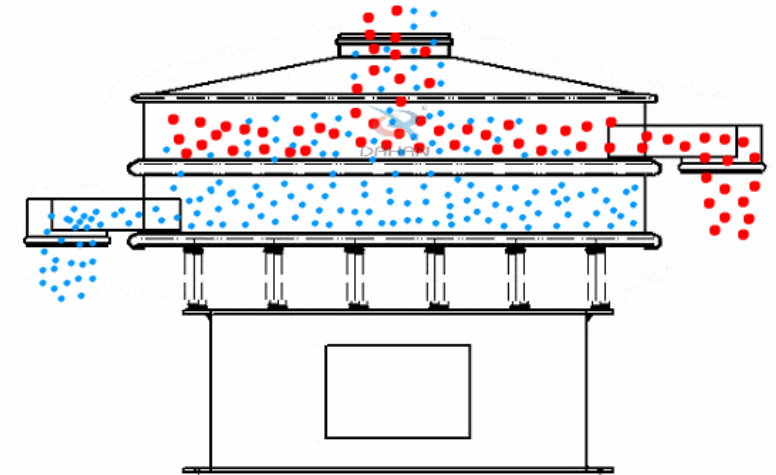
Gyratory Screens

Basic Mechanism

- **Gyratory motion** imparted to a slightly inclined screen casing
- Motion is **circular or elliptical** in a **vertical plane**
- Produces both **vertical throw** (for stratifying particles) and **horizontal throw** (for moving material along the screen)
- Unlike vibratory screens (high speed, small amplitude), gyrating screens use **moderate speed (600–1,800 r/min)** with **larger amplitude**

Working Principle

1. **Feed Entry:** Material is dropped near the **highest point** of the inclined screen
2. **Stratification:** Gyration causes particles to bounce and roll. Fine particles migrate **downward** through the bed of material. Coarse particles rise to the top
3. **Size Separation:** Fines contact the screen surface and pass through openings. Oversize particles continue moving toward the discharge end
4. **Multi-Deck Action:** Material falls to lower decks if present. Each deck produces a different size fraction



Gyratory Screens

Construction Details

1. **Frame and Casing:** Heavy-duty steel casing, Inclined at **16° to 30°** from horizontal
2. **Screen Decks:** Two or more decks stacked vertically (commonly 2 to 5 decks), Rectangular shape, Typical sizes: from 1.5 ft × 4 ft up to 5 ft × 14 ft
3. **Drive Mechanism:** Eccentric shaft mounted **halfway between feed point and discharge**, Produces gyration in a vertical plane about a horizontal axis
4. **Anti-Blinding System:** Rubber balls held in compartments between two screen cloths, During operation, balls bounce and strike the screen surface, Function: Dislodge particles wedged into openings

Applications in Chemical Engineering

- Fertilizer Industry
- Plastic Recycling
- Mineral Processing
- Food Industry
- Chemical Powders

Limitations

- **Not suitable for wet screening** – Gyration motion does not effectively separate wet or sticky materials
- **Poor performance with fibrous materials** – Fibers tend to mat on the screen surface
- **Requires free-flowing feed** – Materials with poor flow properties will not stratify properly
- **Higher maintenance than stationary screens** – Eccentric drive and bearings require regular attention

Material Balance Over a Screen

To determine the flow rates of **overflow (oversize)** and **underflow (undersize)** streams from screening data.

Nomenclature:

F = Feed flow rate , **D** = Overflow (Oversize) flow rate, **B** = Underflow (Undersize) flow rate

Mass fraction of oversize material: **x_F** = in feed, **x_D** = in overflow, **x_B** = in underflow

Material Balance Equations:

Overall Mass Balance

$$F = D + B$$

Oversize Material Balance

$$F x_F = D x_D + B x_B$$

Fraction reporting to overflow:

$$\frac{D}{F} = \frac{x_F - x_B}{x_D - x_B}$$

Fraction reporting to underflow:

$$\frac{B}{F} = \frac{x_D - x_F}{x_D - x_B}$$

- Material balance allows prediction of **oversize and undersize flow rates** using screen analysis data.
- These equations form the basis for evaluating **screen performance and separation efficiency**.

Agitation

What is Agitation?

Agitation is the induced motion of a material in a specified manner, usually producing a circulatory flow pattern inside a vessel.

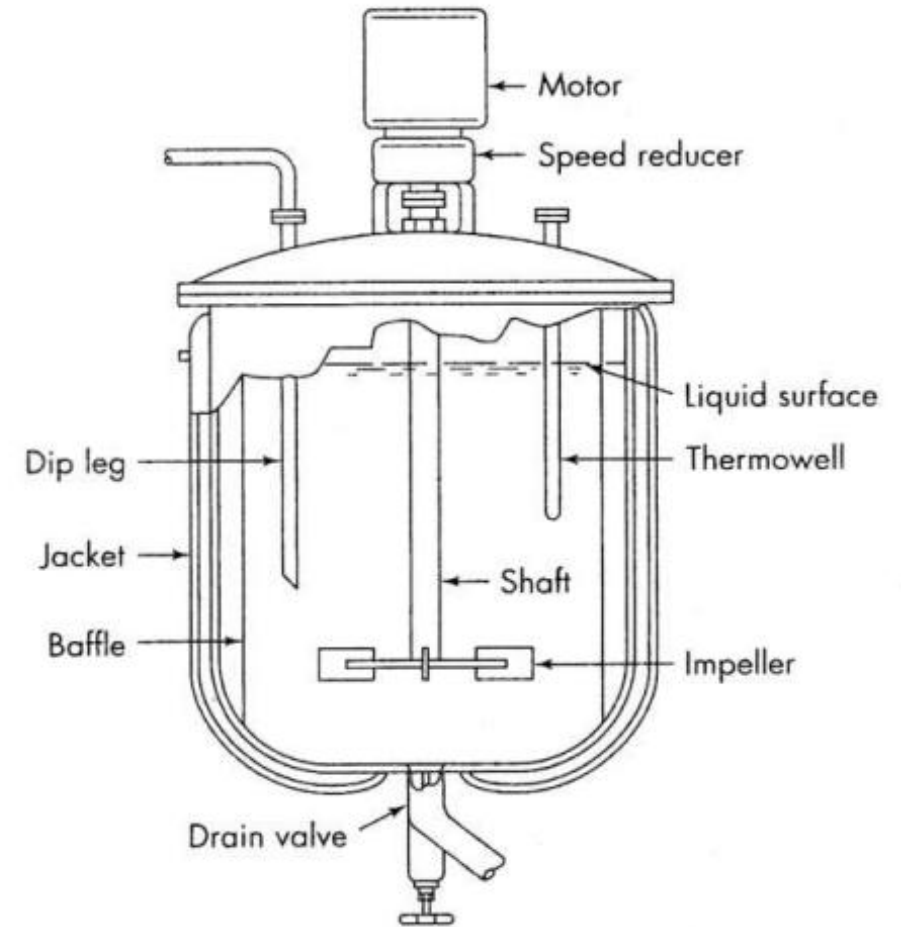
Example: Stirring water in a tank with an impeller.

What is Mixing?

Mixing is the random distribution of two or more initially separate phases into and through one another.

Examples:

- Hot water + cold water
- Powder + liquid
- Gas + liquid



Agitation setup

AGITATED VESSEL AND ITS COMPONENTS

An **agitated vessel** is a tank equipped with an impeller to induce fluid motion and achieve mixing, suspension, gas dispersion and heat transfer.

Key Features

- Cylindrical vessel with vertical axis
- Top may be open or closed
- Rounded (dished) bottom – eliminates dead zones and ensures better circulation
- Liquid depth $\approx$ tank diameter
- Impeller mounted on an overhung shaft driven by a motor (through gearbox, usually)
- Baffles reduce tangential motion and improve mixing
- Accessories: inlet/outlet lines, jackets or coils for heating/cooling, thermowell for temperature measurement, drain valve, etc.



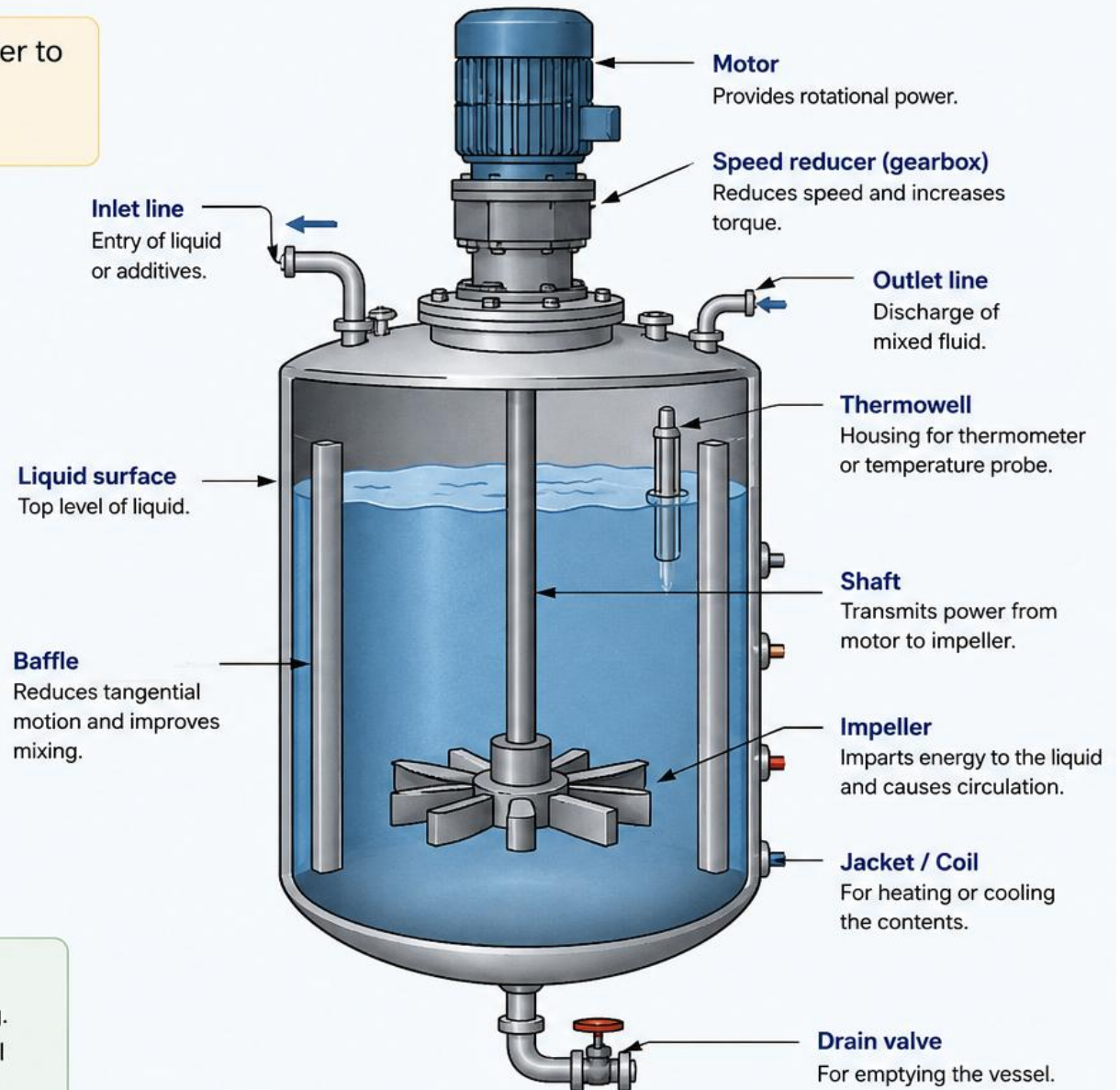
Function

The impeller imparts energy to the liquid causing it to circulate through the vessel and return to the impeller, producing effective mixing.



Why Rounded Bottom and Baffles?

- Rounded bottom prevents stagnant regions and poor mixing.
- Baffles break rotational flow (vortex) and convert it into axial circulation and turbulence for effective mixing.



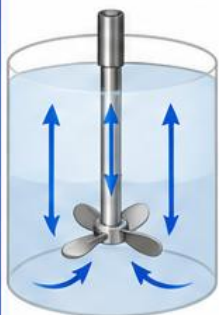
IMPELLERS FOR AGITATION

An **impeller** is the rotating element of an agitator that transfers mechanical energy to the fluid and produces mixing and circulation.

CLASSIFICATION BASED ON FLOW PATTERN

AXIAL-FLOW IMPELLERS

Fluid moves parallel to the shaft

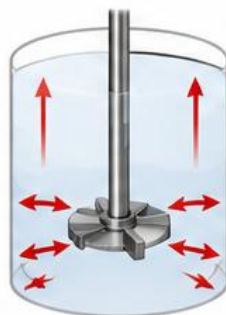


- Top-to-bottom circulation
- High pumping capacity
- Low power consumption

Examples: Propeller, Pitched Blade Turbine, Hydrofoil (A310)

RADIAL-FLOW IMPELLERS

Fluid moves perpendicular to the shaft



- High shear generation
- Strong turbulence
- Excellent gas dispersion

Examples: Straight-Blade Turbine, Disk (Rushton) Turbine, CD-6 Turbine

COMMON IMPELLERS FOR LOW- TO MODERATE-VISCOSITY LIQUIDS

(a)

Marine Propeller



- Axial flow
- High speed (400–1750 rpm)
- Low viscosity liquids

(b)

Straight-Blade Turbine (Paddle)



- Radial flow
- No vertical flow
- 20–150 rpm
- General mixing

(c)

Disk (Rushton) Turbine



- High shear zones
- Excellent for gas dispersion

(d)

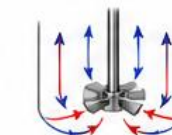
CD-6 Concave Blade Turbine



- Improved gas dispersion
- High efficiency

(e)

Pitched-Blade Turbine



- Axial + radial flow
- Good overall circulation

SELECTION BASED ON FLUID VISCOSITY

Low Viscosity
(water, solvents)



Propeller

- High speed
- Strong circulation
- Effective in large vessels

Moderate Viscosity
(process liquids)



Turbines

- General-purpose mixer
- Good blending
- Gas dispersion

High Viscosity
(polymers, pastes)



Helical Ribbon

- Wall-to-wall mixing
- Up to 25,000 Pa-s

Anchor

- Excellent heat transfer
- Wall scrapers

HIGH-EFFICIENCY IMPELLERS



HE-3 Impeller

- Three slanted blades
- Crimped near tip
- Better axial flow



A310 Fluid-Foil Impeller

- Airfoil-shaped blades
- Tapered blades
- High efficiency
- Lower power

IMPELLERS FOR HIGHLY VISCOUS LIQUIDS

(a) Helical-Ribbon



- Very effective mixing
- Diameter $\approx$ tank diameter
- Up to 25,000 Pa-s

(b) Anchor



- No vertical motion
- Good heat transfer
- Often with wall scrapers



The choice of impeller depends on **viscosity**, desired **flow pattern**, **shear requirement** and **process objective**.

Impellers

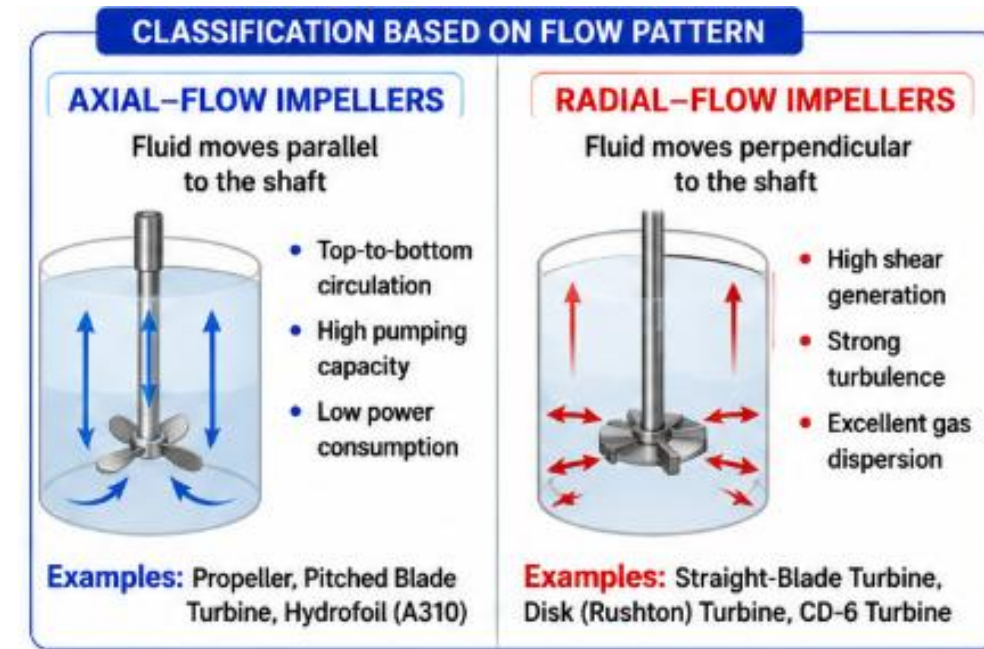
Two Fundamental Classes of Impellers: The way an impeller moves fluid defines its class and determines its application.

1. Axial-Flow Impellers

- Generate currents that flow **parallel** to the axis of the impeller shaft.
- These impellers push fluid in a direction along the shaft, creating a top-to-bottom or bottom-to-top circulation pattern. **Primary Example:** Propellers.

2. Radial-Flow Impellers

- Generate currents that flow in a **radial** or **tangential** direction.
- These impellers push fluid outward toward the vessel walls, perpendicular to the shaft.
- **Primary Examples:** Straight-blade turbines and disk turbines.
- The choice between axial and radial flow is one of the first and most important decisions in agitator selection.



Impeller Selection by Viscosity

For Low- to Moderate-Viscosity Liquids:

Three main types of impellers are used:

- **Propellers** – High-speed, axial-flow impellers.
- **Turbines** – Moderate-speed impellers available in various blade configurations.
- **High-Efficiency Impellers** – Advanced designs that improve flow uniformity and reduce power consumption.

For Very Viscous Liquids:

Standard turbines lose effectiveness as viscosity increases. For these challenging fluids, specialized impellers are required:

- **Helical Impellers** – Designed to sweep the entire tank volume.
- **Anchor Agitators** – Ideal for promoting heat transfer near the vessel wall.



Propeller- Axial-Flow

Design and Speed:

- Propellers are **axial-flow, high-speed** impellers specifically for low-viscosity liquids.
- Small propellers operate at full motor speed: **1,150 or 1,750 rpm**.
- Larger propellers typically turn at **400 to 800 rpm**.

Flow Direction and Effect:

- Rotation is usually chosen to force liquid **downward**.
- The flow currents leaving the impeller continue their path until deflected by the floor of the vessel.
- The highly turbulent, swirling column of liquid leaving the propeller **entrains stagnant liquid** as it moves along, vigorously cutting or shearing the liquid.

Special Configurations:

- In deep tanks, two or more propellers may be mounted on the same shaft, usually directing liquid in the same direction.
- Because of the persistence of flow currents, propellers remain effective even in **very large vessels**.



Turbines – for Moderate Viscosity

Four Types and Their Specific Functions

- Turbines operate at moderate speeds, typically 20 to 150 rpm, and are suitable for a wide range of mixing tasks.

1. Straight-Blade Turbine (Paddle)

- Pushes liquid **radially and tangentially** with almost no vertical motion at the impeller.
- Currents travel outward to the vessel wall, then flow either upward or downward.
- Simple design for general mixing applications.

2. Disk Turbine

- Features multiple straight blades mounted on a horizontal disk.
- Creates zones of **high shear rate**.
- Especially useful for **dispersing a gas** in a liquid; at moderate speeds, gas is forced radially to the blade tips where high shear breaks it into fine bubbles.

3. Concave-Blade CD-6 Disk Turbine

- A specialized variant also widely used for **gas dispersion**.
- The concave blade shape improves gas handling capacity.

4. Pitched-Blade Turbine

- Blades are set at an angle to the shaft.
- Provides **some axial flow** in addition to radial flow.
- Used when **good overall circulation** is important.

Standard Turbines Design

Dimension	Symbol	Standard Value
Impeller Diameter	D_a	$D / 3$
Liquid Depth	H	D
Impeller Height from Bottom	E	$D / 3$
Baffle Width	J	$D / 12$
Blade Width	W	$D_a / 5$
Blade Length	L	$D_a / 4$

Used to calculate:

1. Power number (N_p) calculations
2. Flow number predictions
3. Mixing performance correlations

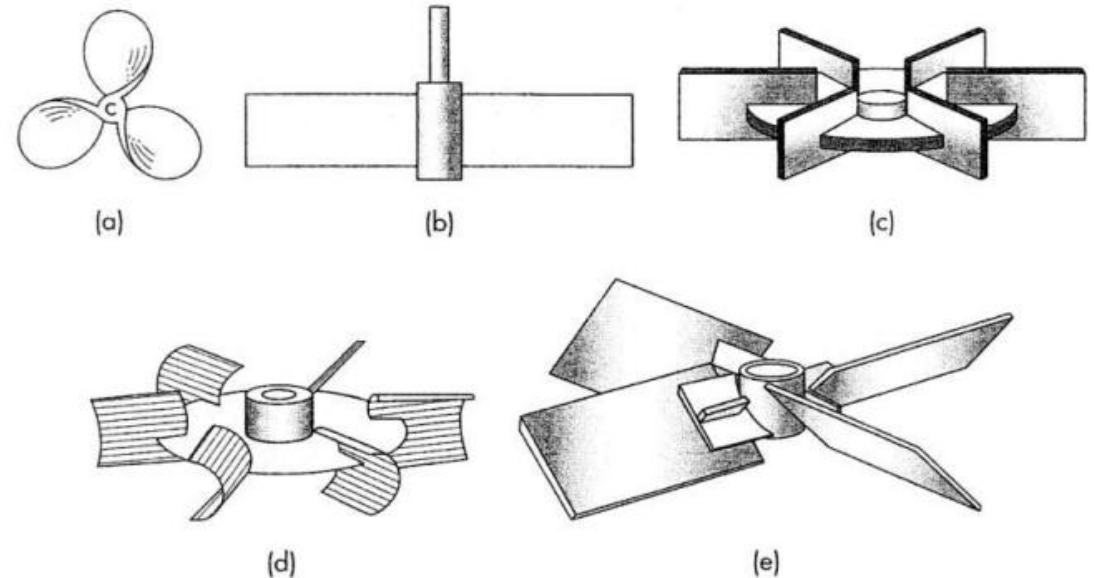


FIGURE 9.2

Impellers for liquids of moderate viscosity: (a) three-blade marine propeller; (b) simple straight-blade turbine; (c) disk turbine; (d) concave-blade CD-6 impeller (Chemineer, Inc.); (e) pitched-blade turbine.

- **Number of baffles:** Usually 4
- **Number of impeller blades:** Ranges from 4 to 16, but generally 6 or 8
- **Industry Example:** used in scale-up calculations from pilot plants to commercial reactors, ensuring that mixing performance is predictable across different scales.

High-Efficiency Impellers

Advanced Designs for Improved Performance

- High-efficiency impellers are variations of the pitched-blade turbine, developed to provide more uniform axial flow, better mixing, and reduced power consumption for a given flow rate.

HE-3 Impeller

- Features three slanted blades that are **crimped** to decrease the blade angle near the tip.
- This design improves flow characteristics and efficiency.

A310 Fluid-Foil Impeller

- Uses **airfoil-shaped blades** which taper so they are narrower at the tip than at their base
- The hydrofoil design minimizes drag while maximizing flow.

Applications and Limitations:

- Widely used to mix **low- or moderate-viscosity** liquids.
- **Not recommended** for very viscous liquids.
- **Not suitable** for dispersing gases.



A310 fluid-foil impeller



HE3 Impeller

Impellers for Highly Viscous Liquids

Helical-Ribbon Impeller

- The helix diameter is very close to the inside diameter of the tank.
- This design **guarantees liquid motion all the way to the tank wall**, even with extremely viscous materials.
- Have been used successfully with viscosities up to **25,000 Pa·s**.
- A double-flight design is commonly used for more effective mixing.



Helical-Ribbon Impeller

Anchor Impeller

- Designed to closely match the contour of the tank, providing good agitation near the floor.
- Creates **no vertical motion**, making it a less effective mixer than a helical ribbon.
- Primary advantage is promoting **heat transfer** to or from the vessel wall.
- Both anchors and helical ribbons may be equipped with **scrapers** that physically remove liquid from the tank wall, further enhancing heat transfer.
- For viscosities above 20 Pa·s, helical-ribbon impellers are often more effective than turbines. Below 50 Pa·s, well-designed turbine systems can still be used.



Anchor Impeller

Flow Patterns in Agitated Vessels

What Determines How Liquid Moves in a Tank?

- **Type of Impeller** – Axial-flow vs. radial-flow designs create fundamentally different initial flow directions.
- **Characteristics of the Liquid** – Viscosity is the most important property; high-viscosity liquids dampen turbulence and change flow behavior.
- **Tank Size and Proportions** – The ratio of liquid depth to tank diameter affects circulation paths.
- **Baffle Configuration** – The presence, number, and size of baffles dramatically alter flow patterns.
- **Impeller Size and Position** – Diameter, blade angle, and height from the tank bottom all influence how fluid circulates.

Every point in an agitated liquid experiences motion that can be broken down into three distinct velocity components. These components work together—or against each other—to create the overall flow pattern.

1. Radial Component

- Acts in a direction **perpendicular to the shaft** of the impeller.
- Moves liquid outward from the impeller toward the vessel walls.
- This component is **useful** for mixing as it pushes fluid into new regions.

2. Longitudinal Component

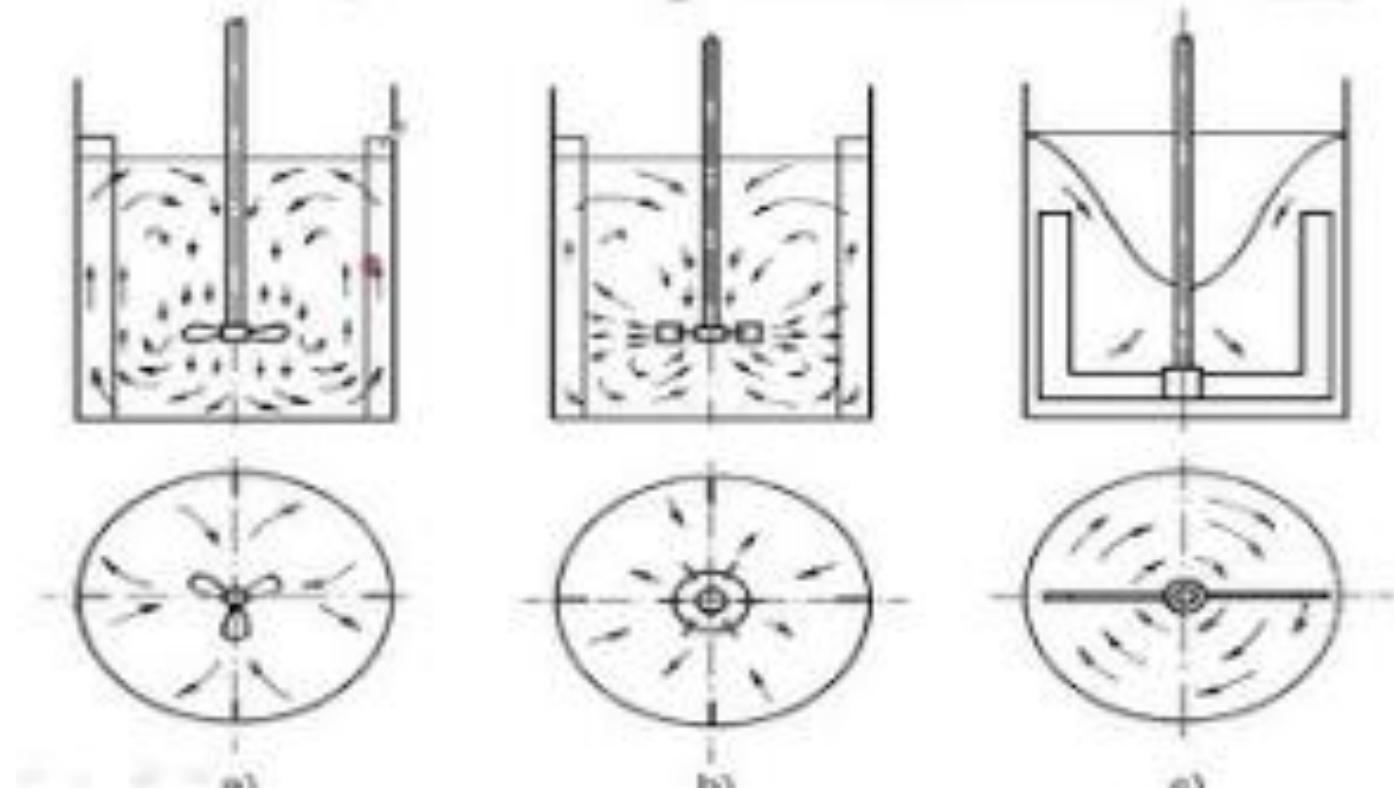
- Acts in a direction **parallel with the shaft**.
- Provides vertical motion, carrying liquid upward or downward along the shaft.
- This component is **essential** for mixing between different levels of the tank.

3. Tangential (Rotational) Component

- Acts in a direction **tangent to a circular path** around the shaft.
- Causes liquid to rotate around the shaft rather than moving inward or outward.
- This component is generally **disadvantageous** in a centrally located, vertical shaft configuration

Radial, Longitudinal, and Tangential Flow Patterns

- a – axial-flow pattern, baffled vessel,
- b – radial-flow pattern, baffled vessel,
- c – tangential-flow pattern, unbaffled vessel



Swirling, Vortex Formation, and Its Consequences

The tangential velocity component creates rotational flow around the shaft. When the shaft is vertical and centrally located, this tangential flow is almost always undesirable.

What Happens During Swirling:

- The liquid follows a circular path around the shaft.
- A **vortex** forms at the liquid surface,
- This swirling pattern is observed with **all types of impellers**—flat-blade turbines, pitched-blade turbines, and propellers—when baffles are absent.

Negative Consequences of Swirling:

- **Stratification is perpetuated** – Fluid at different levels remains separate because there is no longitudinal flow between levels.
- **Reverse concentration occurs** – If solid particles are present, centrifugal force throws them outward. They then move downward and return to the center at the bottom. Instead of mixing, this creates concentration gradients.
- **Reduced relative velocity** – Liquid flows in the same direction as the impeller blades, so the relative velocity between blades and liquid decreases. This **limits the power that the impeller can absorb**.
- **Gas entrainment** – At high speeds, the vortex may deepen until it reaches the impeller, drawing gas from above the liquid down into the charge. This is generally **undesirable**.

Comparing Baffled vs. Unbaffled Vessels

The presence or absence of baffles fundamentally changes the flow pattern and the effectiveness of the mixing process.

In an Unbaffled Vessel:

- **Circulatory (swirling) flow is induced** by all types of impellers, whether axial-flow or radial-flow.
- The flow pattern is virtually the **same regardless of impeller design** when swirling is strong.
- The impeller's intended flow direction (axial or radial) is overwhelmed by the tangential motion.
- A deep vortex forms, leading to stratification, reduced power absorption, and potential gas entrainment.

In a Baffled Vessel:

- Baffles interrupt the tangential flow, converting it into more useful radial and longitudinal motion.
- The intended flow pattern of the impeller is preserved and enhanced.
- **Effective mixing** occurs because fluid is continuously moved between different regions of the tank.
- Power consumption increases because the impeller works against the resistance created by the baffles, improving energy transfer to the liquid.

Prevention of Swirling

Swirling and vortex formation can severely reduce mixing efficiency. Three primary methods are used to prevent this:

Method 1: Off-Center Mounting (For Small Tanks)

The impeller shaft is moved away from the tank's centerline.

The shaft is then tilted in a plane perpendicular to the direction of the move.

This disrupts the symmetrical rotational flow.

Method 2: Side-Entering Mounting (For Larger Tanks)

The agitator is mounted on the side of the tank.

The shaft is positioned in a horizontal plane but at an angle to a radius.

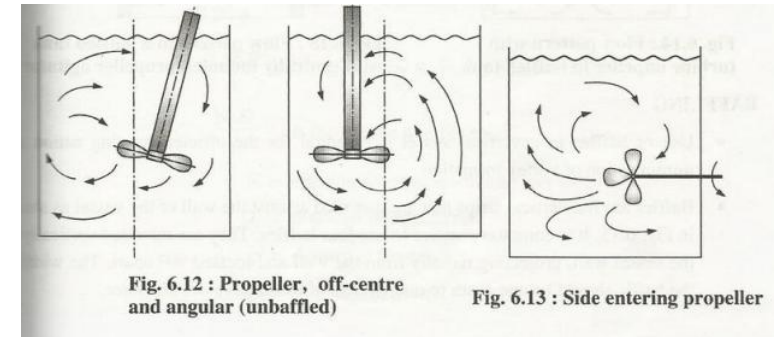
This creates an asymmetric flow that prevents a stable vortex.

Method 3: Installing Baffles (Standard for Large, Vertical Tanks)

This is the most common and effective method.

Vertical strips are installed perpendicular to the tank wall.

They impede rotational flow while allowing radial and longitudinal flow to continue.



Baffle Design & Specifications

Optimizing Baffle Performance

Number of Baffles:

- **Four baffles** are sufficient for most tanks to prevent swirling and vortex formation.
- Even **one or two** baffles have a strong effect on circulation patterns if more cannot be installed.

Baffle Width (Based on Vessel Diameter, D):

- **For Turbines:** Width should be no more than $D/12$.
- **For Propellers:** Width should be no more than $D/18$.

Special Cases (Where Baffles are NOT Required):

- **For highly viscous liquids** (viscosity $> 10 \text{ Pa}\cdot\text{s}$), baffles are not needed as the fluid's own viscosity dampens swirling.
- **For side-entering, inclined, or off-center propellers**, baffles are also unnecessary due to the already asymmetric flow patterns.

Multi-Impeller Systems & Draft Tubes

Designing for Deep Tanks & Specific Applications

Multi-Impeller Systems (For Deep Tanks):

- In a vertical cylindrical tank, the liquid depth should be equal to or somewhat greater than the tank diameter.
- If greater depth is required, two or more impellers are mounted on the same shaft.
- **Standard Configuration:**
 - The **lowest impeller** is commonly a radial-flow unit (e.g., a straight-blade turbine).
 - The **upper impellers** are usually axial-flow units.
 - The lowest impeller is mounted about **one impeller diameter** above the bottom of the tank.

Draft Tubes (For Controlled Flow):

- Typically, return flow to an impeller approaches from all directions (like a fan in a room).
- Draft tubes are used when the direction and velocity of this suction flow need to be controlled.
- **Purpose:**
 - Useful when high shear at the impeller is desired (e.g., manufacturing certain emulsions).
 - Useful for dispersing solid particles that tend to float on the liquid surface.
- **Installation:**
 - For propellers, the draft tube is mounted **around** the impeller.
 - For turbines, it is mounted **immediately above** the impeller.
- **Trade-off:** Draft tubes add fluid friction, reducing the flow rate for a given power input. They are used only when specifically required.

Draft Tubes

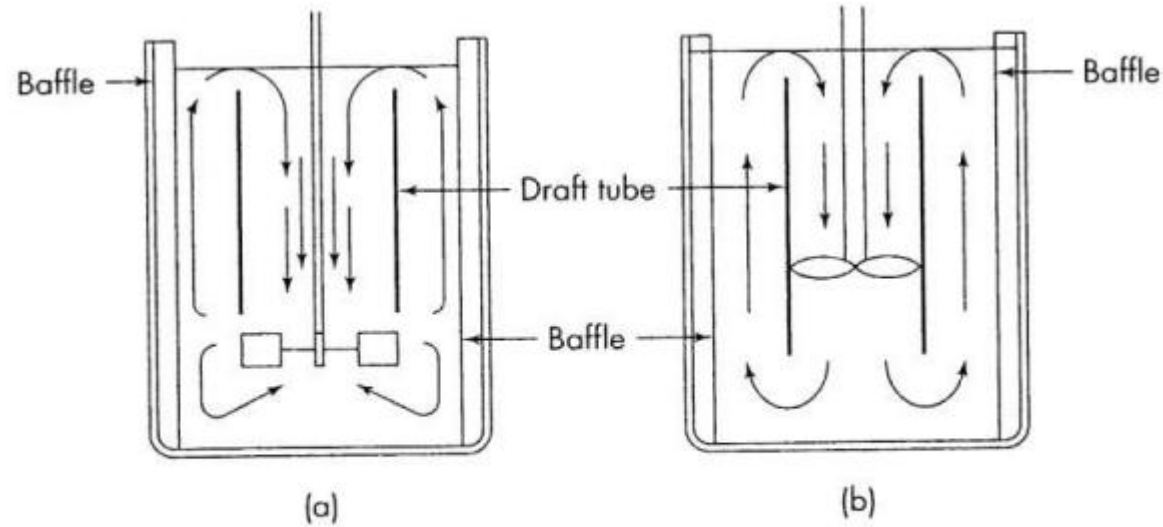


FIGURE 9.8

Draft tubes, baffled tank: (a) turbine; (b) propeller. (After Bissell et al.⁵)

Flow Number (N_Q)

An agitator impeller is essentially a pump impeller operating without a casing. It is a constant that characterizes an impeller's pumping capacity. It represents the discharge flow at the impeller tips.

Key Concept:

The volumetric flow rate (q) from the impeller is proportional to the impeller speed and the cube of its diameter (D_a):

$$q \propto nD_a^3$$

Flow Number Definition:

The Flow Number (N_Q) is the dimensionless ratio that captures this relationship:

$$N_Q = \frac{q}{nD_a^3}$$

Higher N_Q = more flow per rotation.

N_Q for Common Impeller Types:

Marine Propeller (Square Pitch): $N_Q = 0.5$

Four-Blade 45° Pitched Turbine: $N_Q = 0.87$

Disk Turbine (Flat-Blade): $N_Q = 1.3$

HE-3 High-Efficiency Impeller: $N_Q = 0.47$

Power Consumption in Agitated Vessels

Estimating Power in Turbulent Flow

Key Assumption: In turbulent flow, the power (P) required by the impeller can be estimated from the product of:
The flow rate (q) produced by the impeller.
The kinetic energy (E) per unit volume of the fluid.

Flow Equation: $q \propto nD_a^3$

(where n = rotational speed, D_a = impeller diameter)

Kinetic Energy: $E \propto \frac{\rho V_t^2}{2}$

(where ρ = fluid density, V_t = characteristic velocity, slightly less than impeller tip speed)

Resulting Relationship: By combining these, power is found to be proportional to:

$$P \propto \rho n^3 D_a^5$$

This leads to the dimensionless **Power Number (N_p)**, a cornerstone of agitator design.

Power Number (N_p)

Definition: The Power Number is the dimensionless form of the power equation.

Equation (SI Units):

$$N_p = \frac{P}{\rho n^3 D_a^5}$$

Equation (FPS Units - with g_c):

$$N_p = \frac{P g_c}{\rho n^3 D_a^5}$$

Physical Meaning: N_p is analogous to a **friction factor** or a **drag coefficient**.

It represents the ratio of drag force on the impeller to the inertial stress of the fluid.

Example (Standard 6-Blade Turbine):

- The constant N_0 for this impeller is 1.3.
- With an experimentally determined velocity ratio (α), the predicted N_p is approximately **5.8**.
- This theoretical value aligns well with experimental observations.

Final Power Correlation

The Dimensionless Groups: After dimensional analysis, the power function is reduced to a relationship between dimensionless numbers:

$$N_p = \phi(R, e, F, r, S_1, S_2 \dots S_n)$$

The Three Core Dimensionless Groups:

Group	Name	Significance
$N_p = \frac{P}{\rho n^3 D_a^5}$	Power Number	Analogous to a friction factor. Measures resistance to mixing.
$Re = \frac{\rho n D_a^2}{\mu}$	Reynolds Number	Measures ratio of inertial to viscous forces. Dictates flow regime.
$Fr = \frac{n^2 D_a}{g}$	Froude Number	Measures ratio of inertial to gravitational forces. Important for surface vortex formation.

1. The Reynolds Number (Re)

It is proportional to $\frac{\text{Impeller Tip Speed} \times D_a \times \rho}{\mu}$.

Flow Regimes:

- $Re < 10$: Flow is laminar.
- $Re > 10^4$: Flow is fully turbulent.
- $10 < Re < 10^4$: Transition region.

2. The Froude Number (Fr)

It accounts for the effect of gravity on the liquid surface.

When is it important?

- **Important:** In **unbaffled** vessels, where a vortex (swirling depression) forms.
- **Not Important (Fr is neglected):**
 - When **baffles** are used (they prevent vortex formation).
 - When $Re < 300$ (viscous forces dominate, suppressing surface waves).

Introduction to Mixing

Equipment Overlap:

Some equipment used for blending liquids may also be used to mix solids

Liquids:

"Well-mixed" = truly homogeneous liquid phase

Random samples of very small size all have same composition

Pastes & Powders:

Product often consists of two or more identifiable phases

Individual particles may be of considerable size

Small random samples will differ markedly in composition

Mixer Types

Tumbling Mixers:

- V-blenders, double-cone blenders, drum tumblers
- Shell rotates, contents tumble and mix
- **Advantages:** Low cost, easy cleaning, complete discharge, gentle handling
- **Disadvantages:** Batch only, demixing during discharge, not suitable for cohesive materials

Convective Mixers:

- Ribbon blenders, paddle blenders, plowshare mixers
- Stationary vessel with rotating impeller
- **Advantages:** Suitable for both batch and continuous, handles cohesive and free-flowing materials, wide capacity range
- **Disadvantages:** Higher utility and maintenance cost, complex quality analysis

Static Mixers:

- Kenics-type, continuous colliding static mixers
- No moving parts - gravity-induced mixing via particle collisions
- **Advantages:** Energy efficient, minimal maintenance, rapid operation, less particle wear
- **Disadvantages:** Must be designed for specific applications

Plowshare Mixers:

- U-shaped channel with rotating shaft carrying plows
- Creates lifting and shear in powder bed
- **Advantages:** Compatible with wide range of densities
- **Disadvantages:** Not suitable for very cohesive solids; higher speeds cause attrition

Mixers for Noncohesive Solids

Introduction to Dry Powder Mixers

Classification of Solids Mixers:

Some machines used for both heavy pastes AND dry powders

Some machines restricted to free-flowing powders only

Mixing Mechanisms for Dry Powders:

1. **Slow-speed agitation** - Impeller-driven mixing
2. **Tumbling** - Rotation of entire vessel
3. **Centrifugal smearing** - High-speed surface interaction
4. **Impact** - Particle-particle and particle-wall collisions

Key Consideration:

Equipment selection depends critically on material flow properties and processing requirements

Ribbon Blender

Construction:

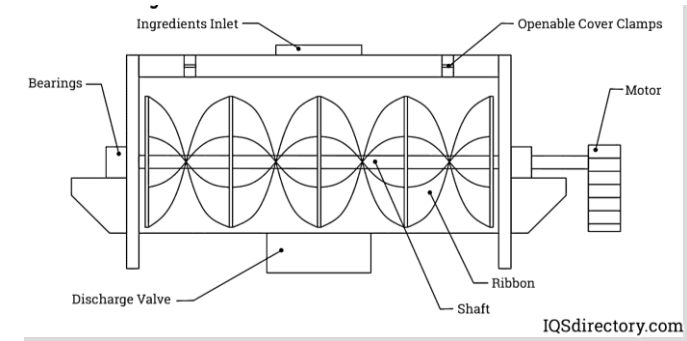
- Horizontal trough-shaped stationary vessel
- Central rotating shaft running the length of the trough
- Two sets of helical ribbons mounted on the same shaft:
 - **Outer ribbon:** Moves material slowly in one direction
 - **Inner ribbon:** Moves material quickly in the opposite direction
- Ribbons may be **continuous** (full-length) or **interrupted** (segmented)

Mixing Mechanism:

- **Convective mixing** – bulk movement of powder by the ribbons
- **Shear mixing** – created by the opposing motion of inner and outer ribbons
- Material is constantly lifted, folded, and sheared between the two ribbon flows
- Creates a **counter-current** action that promotes rapid blending

Operational Flexibility:

- Can operate in **batch mode** – entire batch mixed then discharged
- Can operate in **continuous mode** – feed at one end, discharge at the other



Ribbon Blender

Ribbon Blender – Industrial Use & Constraints

Applications:

- Animal feed production (mixing grains, protein meals, vitamins)
- Food processing (flour blends, spice mixtures)
- Pharmaceutical powder blends (API + excipients)
- Chemical industry (detergents, fertilizers, plastic compounding)
- Construction materials (dry mortar, cement additives)

Limitations:

- High power consumption due to resistance of powder bed
- Not suitable for very cohesive or sticky materials
- May cause particle attrition (breakage) with fragile materials
- Dead zones near trough ends if improperly designed
- Cleaning can be difficult, especially with interrupted ribbons

Tumbling Mixers

General Construction:

- Vessel (drum, cone, or cylindrical shell) rotates about a horizontal axis
- Charged from above through an inlet port
- Rotated at speeds typically 20–50% of critical speed
- May include internal sprays for liquid addition
- May contain mechanical devices to break agglomerates



Types of Tumbling Mixers:

Type	Construction Feature
Ball Mill	Cylindrical shell with grinding media; handles dense slurries and heavy solids
Tumbling Drum	Simple rotating cylinder; handles heavy solids and dense slurries
Double-Cone Mixer	Two conical sections joined at the center; restricted to light free-flowing solids

Tumbling Mixers

Tumbling Mixers – The Problem of Never Complete Mixing

- **Understanding the Fundamental Limitation of Tumbling Blenders**
- **Critical Issue:**
- Mixing is initially rapid, but **never reaches a completely random state**
- After a certain time, mix quality levels off
- Quality may fluctuate or even **decrease** with longer mixing

Why Does This Happen?

- **Unblending forces** are always present in dry solids blenders
- These include:
 - **Electrostatic forces** – particle charging due to friction
 - **Segregation by size** – smaller particles migrate downward
 - **Segregation by density** – denser particles settle, lighter ones rise
- These forces oppose the mixing action and become dominant over time

Practical Implication:

- Extended mixing time does NOT guarantee better product
- Optimal mixing time must be determined experimentally
- Overmixing can lead to significant unmixing and segregation

Impact Wheel Mixers

Impact Wheel Technology – Design & Working

Construction:

- Central high-speed spinning disk, 250–700 mm (10–27 in.) in diameter
- Disk is enclosed in a stationary casing
- Premix of dry ingredients fed continuously near the center of the disk
- Disk rotates at very high RPM, throwing material outward

Mixing Mechanism:

- **Centrifugal smearing** – intense shearing forces as powder travels over the disk surface
- **Impact** – particles collide with the stationary casing and with each other
- High-speed dispersion creates intimate mixing of ingredients
- Combines surface interaction (smearing) with particle collision (impact)

Operation:

- Continuous feed and discharge
- Handles 1 to 25 tons/hour of light, free-flowing powders

Impact Wheel Mixers

Applications:

Pesticide and insecticide powder formulations

Fine chemical powder blending

Dry toner for printing industries

Where continuous high-throughput mixing is required

Light, free-flowing powders requiring intense mixing

Limitations:

Restricted to light, free-flowing powders only

Not suitable for dense, heavy, or cohesive materials

High-speed operation causes significant particle attrition

Heat generation due to impact may affect heat-sensitive materials

Requires premixing before feeding to the disk

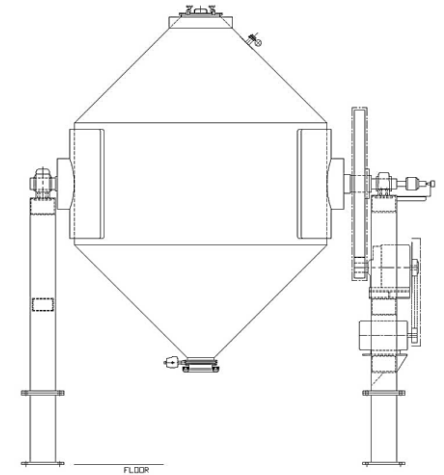
Double-Cone Tumbling Mixer

Construction:

- Two conical shells joined at their bases, forming a symmetrical "V" or double-cone shape
- Rotates about a horizontal axis passing through the center
- Charged from the top (apex of one cone)
- Fill level: 50–60% of total volume
- May have internal spray nozzles or agglomerate-breaking devices

Mixing Mechanism:

- **Tumbling action** – as the vessel rotates, powder cascades from the top to the bottom
- **Convective mixing** – bulk movement of powder due to gravity
- **Diffusive mixing** – random particle motion during cascading
- Material splits and recombines at each rotation
- No internal agitators – mixing is purely by vessel motion



Double-Cone Tumbling Mixer

Double-Cone Mixer – Industrial Use & Constraints

Applications:

- Pharmaceutical tablet blends (API + excipients)
- Dry powder cosmetics and pigments
- Food industry (spices, instant drink mixes)
- Fine chemical powder blending
- Where gentle mixing and complete discharge are critical

Limitations:

Restricted to light, free-flowing solids only

- Cannot handle cohesive or sticky powders effectively
- **Mixing is never complete** – components never reach a fully random state
- Subject to **unblending forces** – electrostatic charge causes demixing
- Mix quality plateaus and may even decrease with prolonged mixing
- Segregation during discharge is a major concern

Mixer Selection

Mixer Type	Construction	Mechanism	Best For	Key Limitation
Ribbon Mixer	Trough + dual ribbons	Convective + shear	Most powders	High power, not for sticky
Double-Cone Tumbler	Rotating cone pair	Tumbling + gravity	Light free-flowing solids	Never complete, segregation
Tumbling Drum / Ball Mill	Rotating cylinder	Tumbling + grinding	Dense slurries, heavy solids	Less precise blending
Impact Wheel	High-speed disk + casing	Smearing + impact	Light free-flowing powders	Attrition, heat generation

Mixers for Cohesive Solids

Key Characteristics of cohesive solids:

- Pastes, plastic materials, rubber
- High viscosity materials
- Specialized mixing equipment required

Why These Materials Are Difficult to Mix

- Resemble liquids — but enormously high viscosity
- Mixing elements cannot generate flow currents
- Instead: shear, fold, stretch, and compress the mass
- Mechanical energy applied directly by moving parts
- Large forces generated → high power consumption

Change-Can Mixers

Change-Can Mixers: For Viscous Liquids & Light Pastes

Applications:

- Food processing
- Paint manufacture
- **Size Range:** 5 to 100 gallons

Two Main Types:

Pony Mixer

- Rotating agitator with vertical blades near vessel wall
- Can rotates in opposite direction to agitator on turn table

Beater Mixer

- Stationary can
- Planetary motion agitator (precesses to visit all parts of vessel)

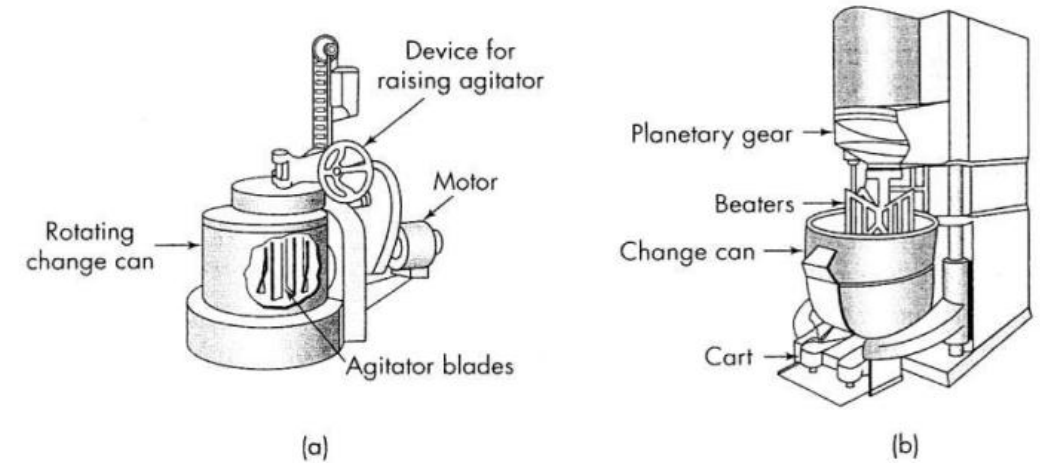


FIGURE 28.3
Double-motion paste mixers: (a) pony mixer; (b) beater mixer.



Kneading Machines

For Deformable & Plastic Solids

Mixing Actions:

- Squashing flat
- Folding over
- Squashing again (repeated)

Additional Mechanisms:

- Tearing the mass apart
- Shearing between moving blade and stationary surface

Power Requirements:

- Considerable energy even for thin materials
- VERY LARGE for stiff, rubbery masses

Types of Kneading Equipment

Kneader

- Handles: Suspensions, pastes, light plastic masses
- Applications: Compounding lacquer bases, shredding cotton linters

Disperser

- Heavier construction
- Higher power than kneader
- Works additives into stiff materials

Masticator (Intensive Mixer)

- Heaviest construction
- Highest power consumption
- Disintegrates scrap rubber
- Compounds toughest plastic masses

Kneading Machines

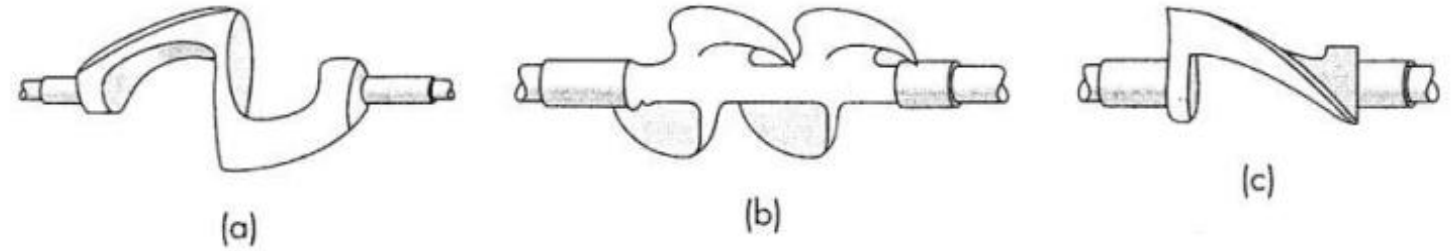


FIGURE 28.4

Kneader and disperser blades: (a) sigma blade; (b) double-naben blade; (c) disperser blade.

Mixer Blade Configurations

Specialized Blades for Different Applications

Sigma Blade

- General-purpose kneading
- Can have serrated edges for shredding action

Double-Naben (Fishtail) Blade

- Effective with heavy plastic materials

Disperser Blade

- High shear forces
- Disperses powders/liquids into plastic/rubbery masses

Masticator Blades

- Even heavier construction
- Spiral, flattened, elliptical designs available

Internal Mixers (Closed Chamber)

Sealed Operation During Mixing Cycle

Key Features:

- Mixing chamber closed with cover
- Cover conforms to volume swept by blades
- Does not tilt

Applications:

- Dissolving rubber
- Making rubber-in-liquid dispersions

Banbury Mixer

The Most Common Internal Mixer:

- Heavy-duty two-arm design
- Interrupted spiral agitators
- Shaft speed: 30-40 rpm

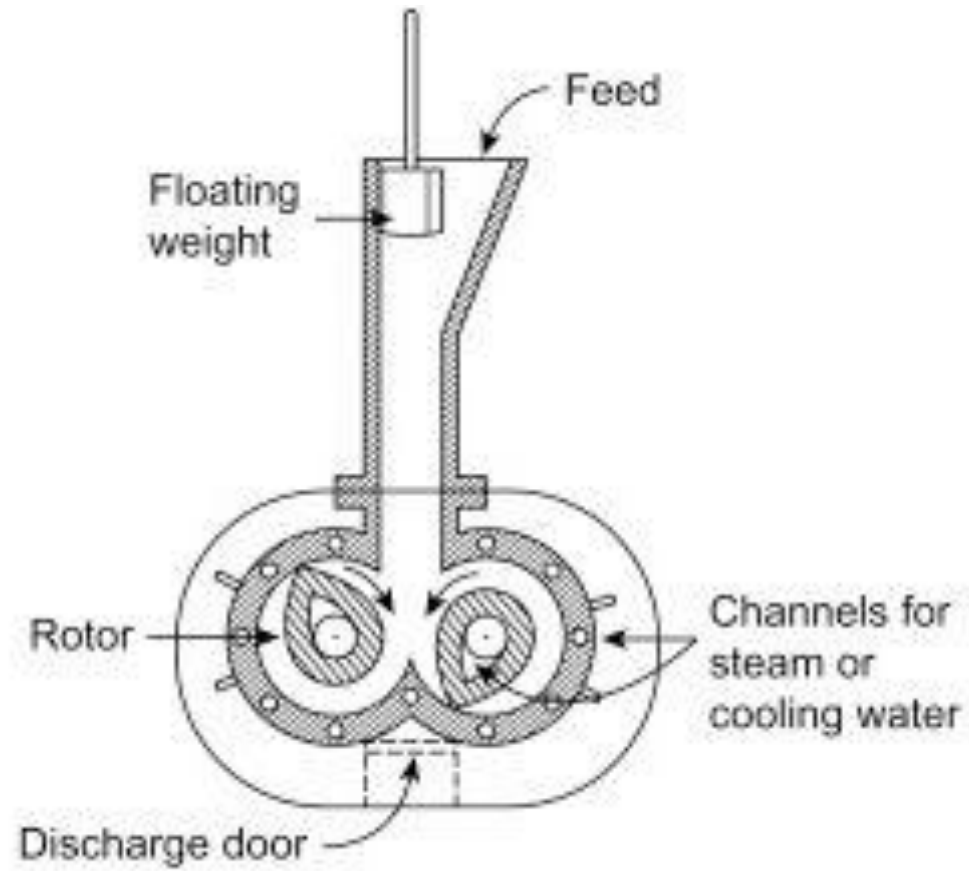
Operation:

- Solids charged from above
- Air-operated piston (1-10 atm pressure)
- Discharge through sliding door in bottom

Applications:

- Compounding rubber & plastics
- Masticating crude rubber
- Devulcanizing rubber scrap
- Making water dispersions & rubber solutions

Banbury mixer



Muller Mixers

Distinct Smearing/Rubbing Action

Mulling Action:

- Similar to mortar and pestle
- Smearing/rubbing over solid layer

Design Variations:

Design Type	Moving Component
Standard	Pan stationary, wheels rotate
Alternative 1	Wheels stationary, pan rotates
Alternative 2	Both pan & wheels driven (offset)

Mulling Action:

- Wide, heavy wheels roll in circular path
- Rubbing from wheel slip on solids
- Plows guide solids under wheels

Applications:

- Heavy solids and pastes
- Coating particles with small amount of liquid

Available:

- Batch units
- Continuous units with two pans in series

Mixing

Banbury mixer

